COURSE SLIDES · WSQ

WSQ Technical Drawing with AutoCAD

WSQ Course Code: TGS-2021005538

Skills Framework TSC: Technical Drawing (DSN-TDR-3005-1.11)

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v13 · 21 August 2026

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COURSE ADMINISTRATION

Welcome & Housekeeping

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Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer/administrator displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

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YOUR TRAINER · TEMPLATE

About the Trainer



Trainer Profile Template

Complete before delivery

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

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YOUR TRAINER

About the Trainer



Dr Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte Ltd

ROLE
Principal Trainer, Tertiary Infotech Academy Pte Ltd

SPECIALISATION
CAD, technical drawing and engineering design — AutoCAD, Revit and Fusion 360.

DELIVERS
WSQ courses on technical drawing, design and emerging technologies.

FOUNDER
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

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ICE-BREAKER

Let's Know Each Other

- Your name and organisation / role.
- Your experience with CAD, drafting or design (if any).
- What you want to be able to draw or document after this course.

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HOUSEKEEPING

Ground Rules

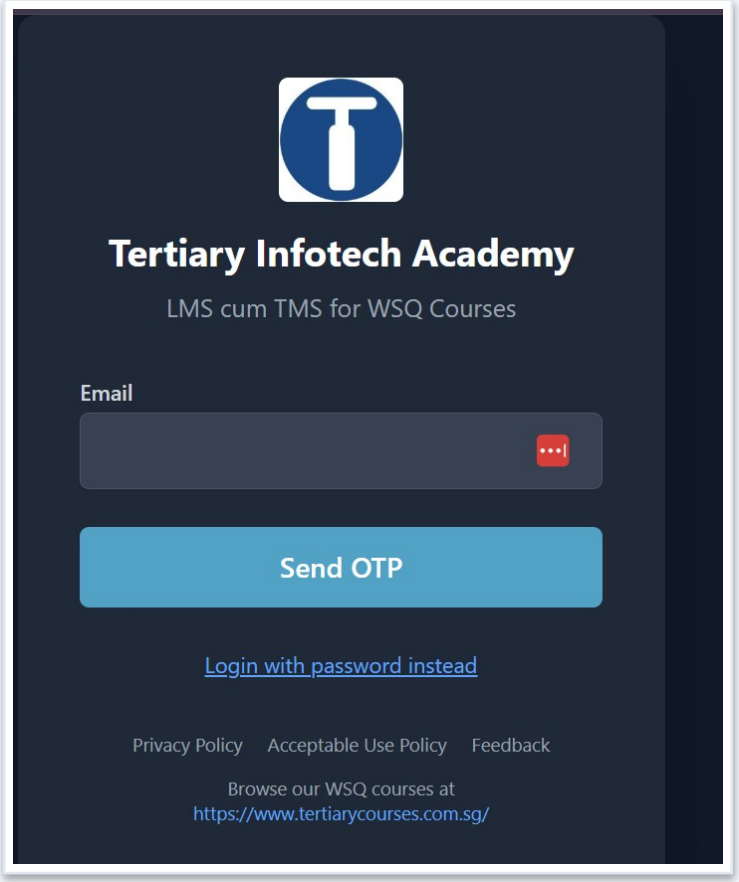
1. Set your mobile phone to silent mode.
2. Participate actively — no question is stupid.
3. Mutual respect: agree to disagree.
4. One conversation at one time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required.

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COURSE PORTAL · LMS/TMS

Download Course Material

- Log in to the LMS/TMS portal with the account created at registration.
- Portal: https://lms-tms.tertiaryinfotech.com
- Open your course → Course Materials to download the slides and Learner Guide.
- Keep them handy — the final assessment is open book (slides, Learner Guide and approved materials only).



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SCHEDULE

Lesson Plan — 2 Days, 8 hours/day

Day 1

- Day 1 — AutoCAD Foundations, Drawing & Modifying Objects
  - Digital attendance (AM) · introductions · outcomes
  - Topic 1: Getting Started with AutoCAD (Lab 1)
  - Topic 2: Technical Drawing with AutoCAD (Labs 2–5)
  - Topic 3: Modifying Objects (Labs 6–8)

Day 2

- Day 2 — Organizing Objects, Design Documentation & Assessment
  - Topic 4: Organizing Objects (Labs 9–12)
  - Topic 5: Design Documentation (Labs 13–16)
  - Course feedback & TRAQOM survey
  - Final Assessment: WA (SAQ) + Practical Performance
  - Daily timing: 9:30am–6:30pm · 1-hour lunch

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WSQ ALIGNMENT

Skills Framework — TSC

1

TSC Title

Technical Drawing

2

TSC Code

DSN-TDR-3005-1.11

3

Framework

Skills Framework for Design — Technical Drawing, Proficiency Level 3.

4

Coverage

Five abilities (A1–A5) and five knowledge areas (K1–K5) mapped to the five course topics.

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SKILLS FRAMEWORK TSC

TSC Abilities & Knowledge

Abilities

- A1: Read and visualise technical drawings according to briefs
- A2: Produce two-dimensional (2D) and three-dimensional (3D) technical drawings based on input from stakeholders
- A3: Create part lists for new design to facilitate technical drawing
- A4: Oversee the development and construction of new designs' prototypes, in collaboration with organisation's production function
- A5: Produce drawings and specifications, that conform to the requirements of relevant standards for submission

Knowledge

- K1: Symbols, standards and conventions used in technical drawings within one's field and region
- K2: Methods of technical drawing by instruments and computer-aided design (CAD) systems
- K3: Types of technical drawings
- K4: Fundamentals of drafting design documentation
- K5: Concepts of information technology (IT) to facilitate the usage of technical drawing software

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WHAT YOU'LL ACHIEVE

Learning Outcomes

1

LO1 — Read

Read and visualize AutoCAD technical drawings.

2

LO2 — Produce

Produce 2D and 3D technical drawings using AutoCAD.

3

LO3 — Part lists

Create part lists using AutoCAD blocks, attributes and tables.

4

LO4 — Document

Create AutoCAD design documentation — text, dimensions, annotation.

5

LO5 — Specify

Produce design specification: layouts, viewports and plotted sheets.

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FIVE TOPICS · 16 HANDS-ON LABS

Course Outline

Day 1

- Topic 1: Getting Started with AutoCAD (A1, K1)
  - AutoCAD user interface
  - Workspace settings, navigation and units
- Topic 2: Technical Drawing with AutoCAD (A2, K2)
  - Basic AutoCAD operations
  - Drawing objects and isometric drawing
- Topic 3: Modifying Objects (A3, K3)
  - Working with modifiers: move, copy, trim, arrays…

Day 2

- Topic 4: Organizing Objects (A4, K4)
  - Object properties and layers
  - Blocks, attributes and reusable content
- Topic 5: Design Documentation (A5, K5)
  - Text, dimensions and part-list tables
  - Layouts, printing and outputs
  - Annotative techniques

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ASSESSMENT RULES

Briefing for Assessment

- Place phones and other materials under the table or on the floor.
- No photos or recording of assessment scripts.
- No discussion during the assessment.
- Use a black/blue pen for hard-copy assessments.
- No liquid paper / correction tape.
- Scripts are collected when time is up.

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Final Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — produce a dimensioned AutoCAD technical drawing, 1.5 hours, open book.
- Format: Open Book — slides, Learner Guide and approved materials only.
- A minimum of 75% attendance (SSG Digital Attendance record) and a 'Competent' assessment result are required for funding.
- An appeal process is available if required.

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ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit WA (SAQ) then Practical Performance — open book
4. Submit your answers and DWG file on the LMS
5. Sign the Assessment Summary Record

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CORE CONCEPTS

Technical Drawing & CAD Fundamentals

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What is a Technical Drawing?

1

Precise communication

A technical drawing communicates a design's exact geometry, sizes and finishes for manufacture or construction.

2

Standards-based

Symbols, line conventions, projections and dimensioning follow standards (ISO/ANSI) so any reader interprets it the same way.

3

Types

Orthographic multi-view, isometric/pictorial, section, detail and assembly drawings — each answers a different question.

4

CAD advantage

AutoCAD makes drawings precise, editable, layered, reusable and printable at any scale.

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METHODOLOGY

The Technical Drawing Workflow

Set up & draw

- Template, units and limits
- Precision aids: snap, ortho, polar
- Draw geometry at 1:1

Organise & refine

- Layers, properties, blocks
- Modify: trim, fillet, array
- Attributes for part data

Document & publish

- Dimensions, text and tables
- Layouts with scaled viewports
- Plot to PDF for submission

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01

TOPIC 01

Getting Started with AutoCAD

AutoCAD User Interface · Workspace Settings · Navigation · Units

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SKILLS FRAMEWORK A1 · K1

Key Concepts — Getting Started with AutoCAD

1

User interface

Application menu, Quick Access Toolbar, ribbon tabs and panels, drawing area, command window and status bar.

2

Workspaces

Drafting & Annotation, 3D Basics and 3D Modeling — task-oriented sets of ribbons, menus and palettes.

3

Model vs paper space

Design at 1:1 in model space; compose scaled drawing sheets in paper-space layouts.

4

Precision aids

Grid and snap, ortho mode, polar tracking, object snaps and dynamic input keep every pick exact.

5

Navigation

Zoom, pan, ViewCube and the wheel mouse move you around a drawing without changing it.

6

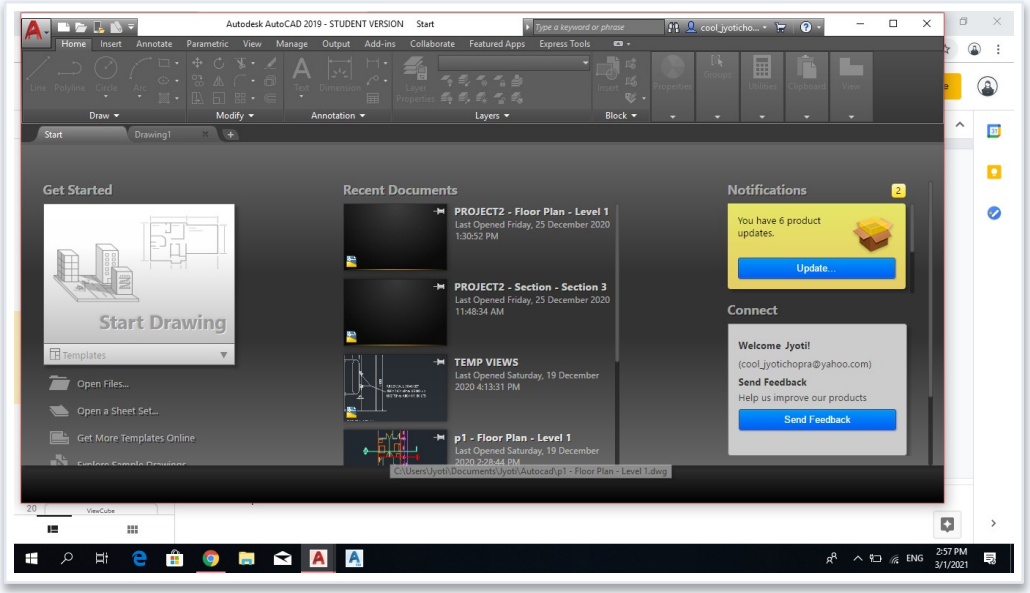
Units and limits

UNITS sets the length/angle format and precision; LIMITS frames the working area of the drawing.

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AutoCAD User Interface — Application Menu

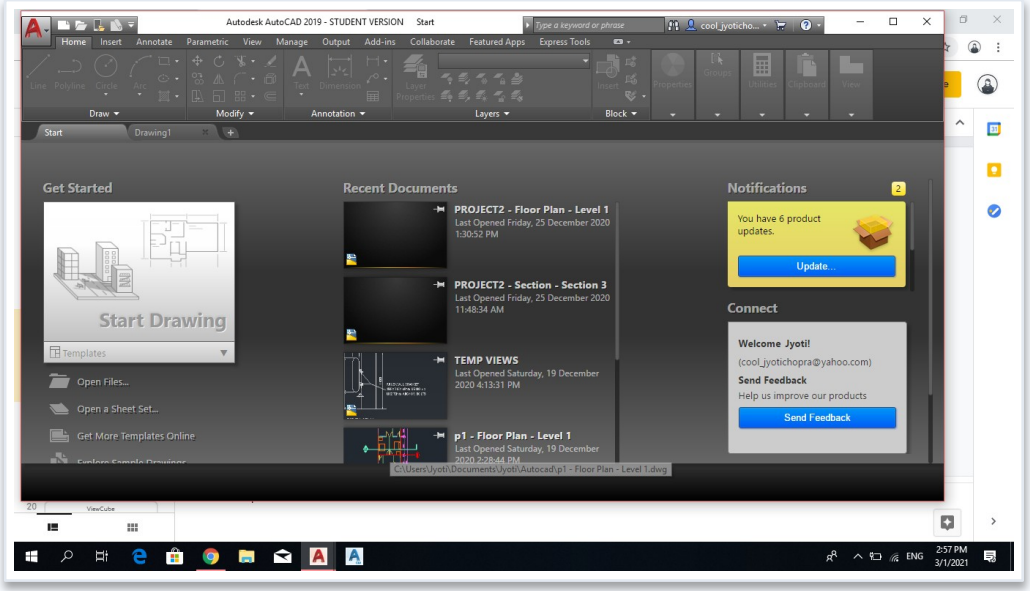
- The Application button opens the file-level menu: create, open and save drawings.
- Audit, recover and purge a drawing file when it needs repair or cleanup.
- Print or publish a file, and access the Options dialog for system settings.
- Recent documents are listed for quick reopening.



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Starting a New Drawing

- Start New Drawing creates a file from the default drawing template (DWT).
- Metric drawings use acadiso.dwt; imperial drawings use acad.dwt.
- The default template is set in the Options dialog.
- Templates carry units, styles, layers and title blocks — start right, draw less.

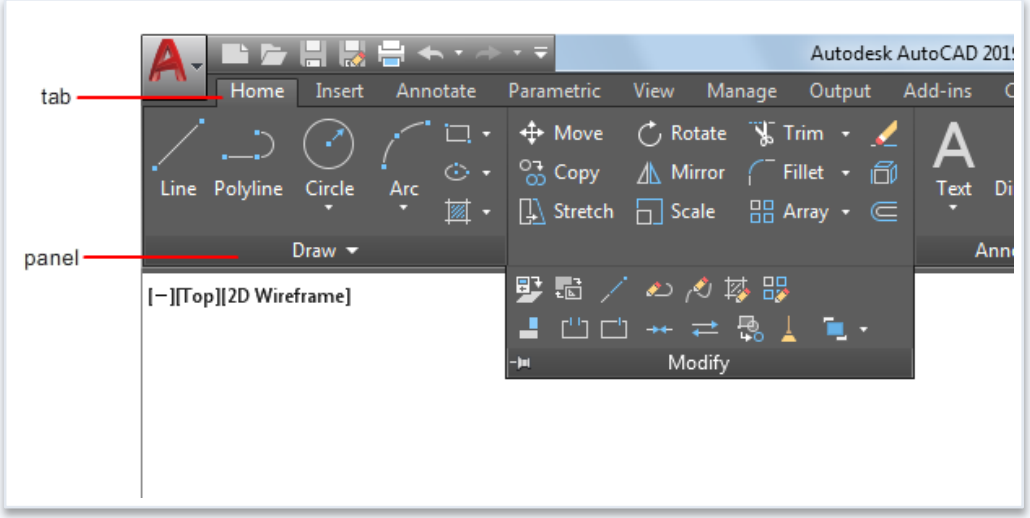


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USER INTERFACE

Ribbon Tabs and Panels

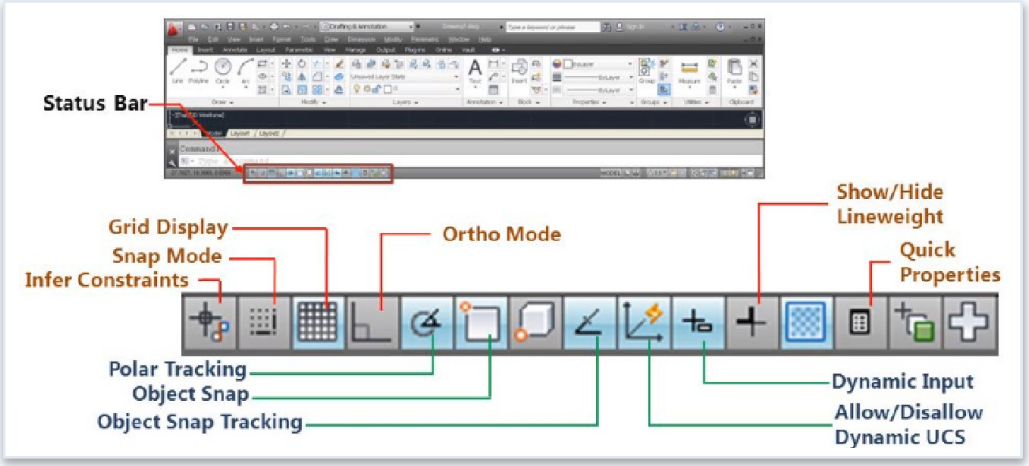
- The ribbon organises tools into tabs (Home, Insert, Annotate, View, Output…).
- Each tab is a set of panels — Draw, Modify, Layers, Annotation and more.
- Panels expand to reveal less-used tools; pin them open while learning.
- Most course commands live on the Home tab's Draw and Modify panels.



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Status Bar — Precision Drafting Aids

- The status bar toggles grid, snap, ortho, polar tracking and object snap.
- Ortho constrains the cursor to horizontal/vertical; Polar tracks set angles.
- 2D object snap picks exact endpoints, midpoints, centres and intersections.
- Function keys mirror the toggles: F7 grid · F8 ortho · F9 snap · F10 polar · F3 osnap.

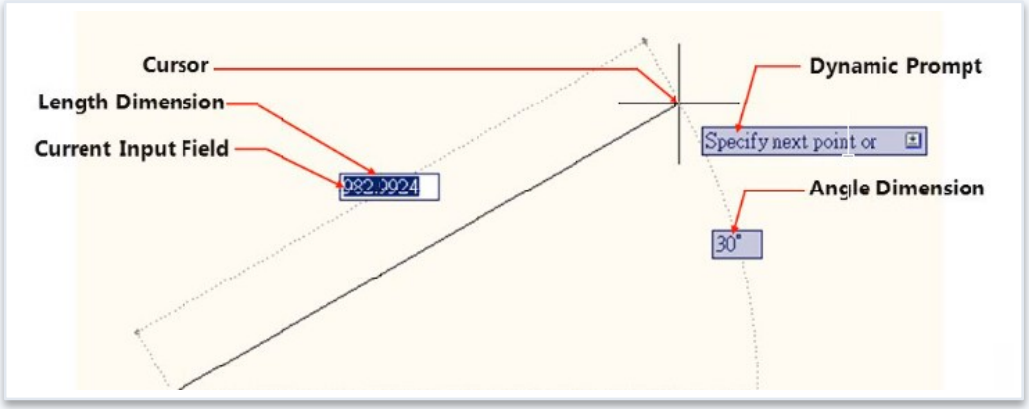


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USER INTERFACE

Dynamic Input and the Command Window

- Dynamic input places a command interface beside the cursor as you draw.
- Tooltips show prompts, dimensions and coordinate entry at the crosshair.
- The command window (Ctrl+9) accepts commands and shows every prompt.
- Watch the prompts — AutoCAD always tells you what it expects next.

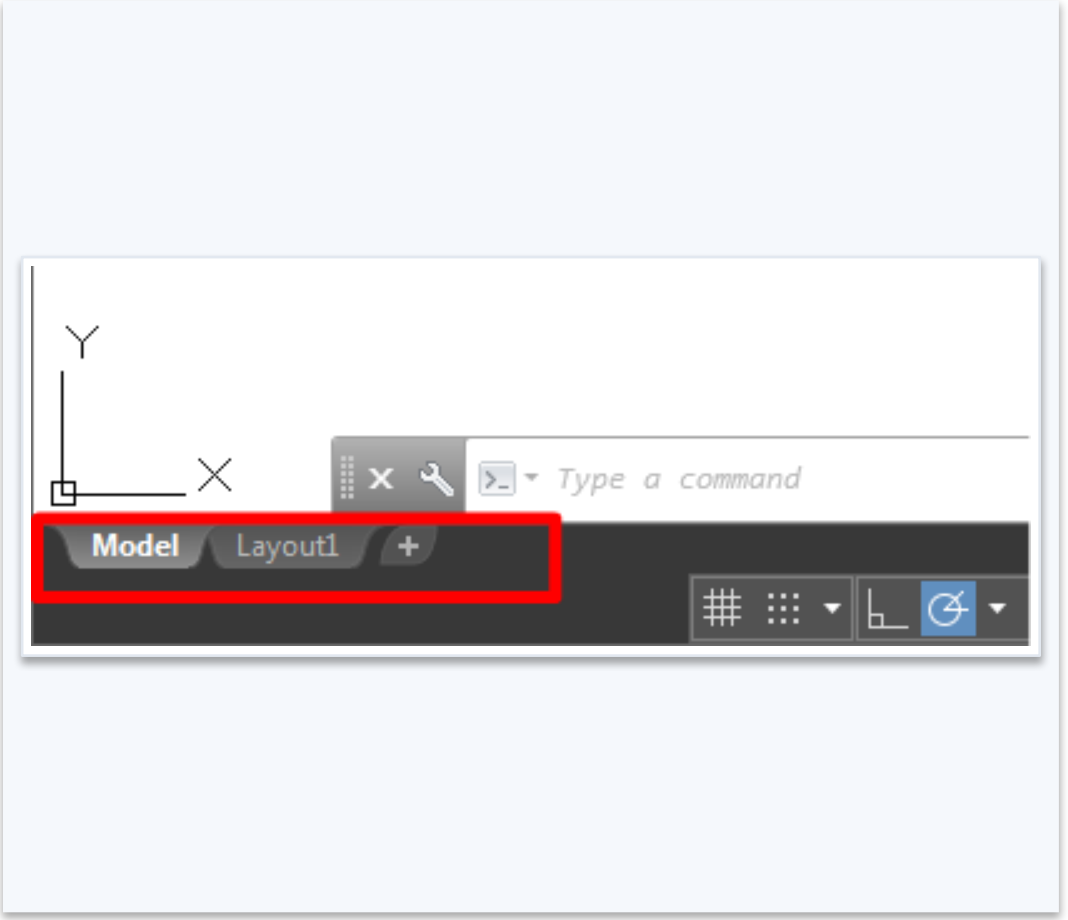


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WORKSPACE

Model Space and Paper Space

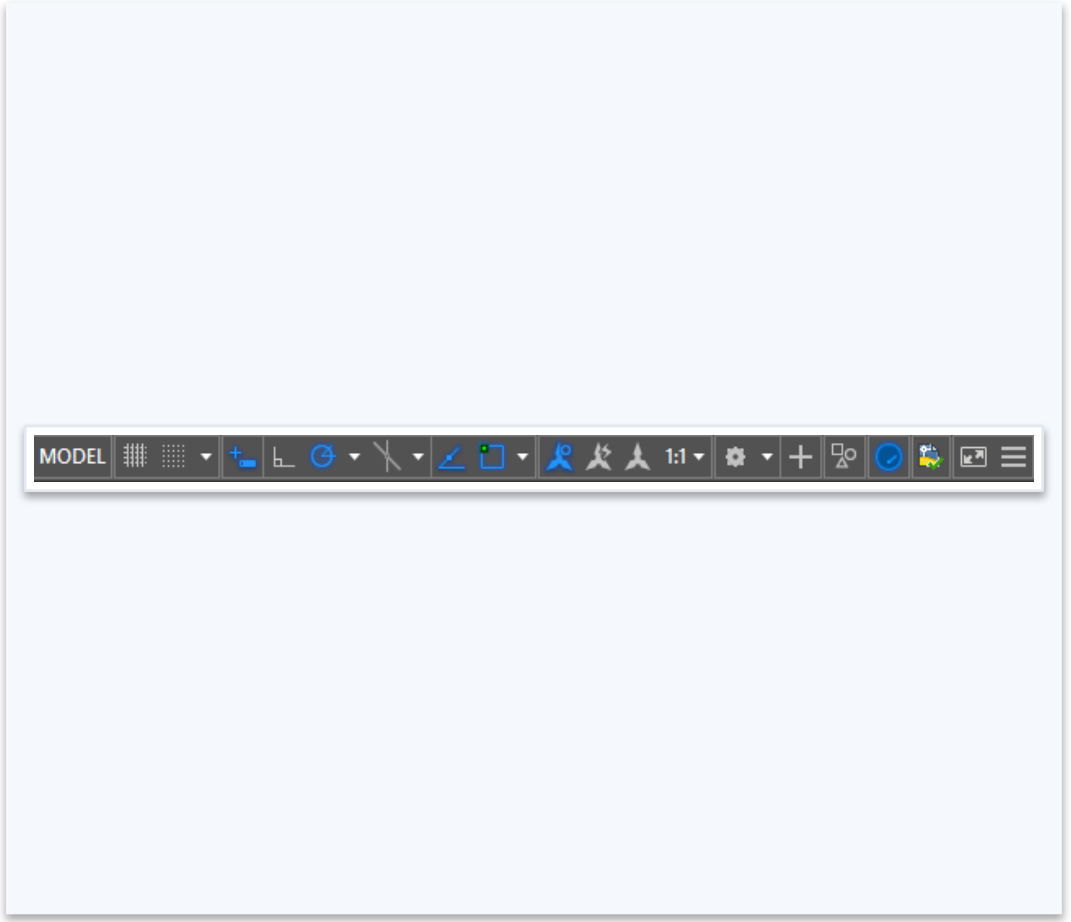
- Model space is where you design — always draw at 1:1 real-world size.
- Paper space (layouts) is where you compose scaled sheets for printing.
- The UCS icon shows the origin and orientation of the coordinate system.
- Switch between them with the Model and Layout tabs below the drawing area.



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Workspaces and the Quick Access Toolbar

- Workspaces group menus, toolbars and ribbons for a task: Drafting & Annotation, 3D Basics, 3D Modeling.
- This course uses Drafting & Annotation throughout.
- The Quick Access Toolbar sits top-left: New, Open, Save, Undo, Plot.
- Right-click any ribbon tool → Add to Quick Access Toolbar to customise it.



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Navigation — Zoom, Pan and the Wheel Mouse

- The navigation bar provides Pan and the full set of Zoom tools.
- ZOOM (Z): Window, All, Extents, Previous, Scale, Object — know them all.
- Wheel mouse: scroll to zoom, press to pan, double-click to zoom extents.
- Navigation changes your view only — it never moves the geometry.

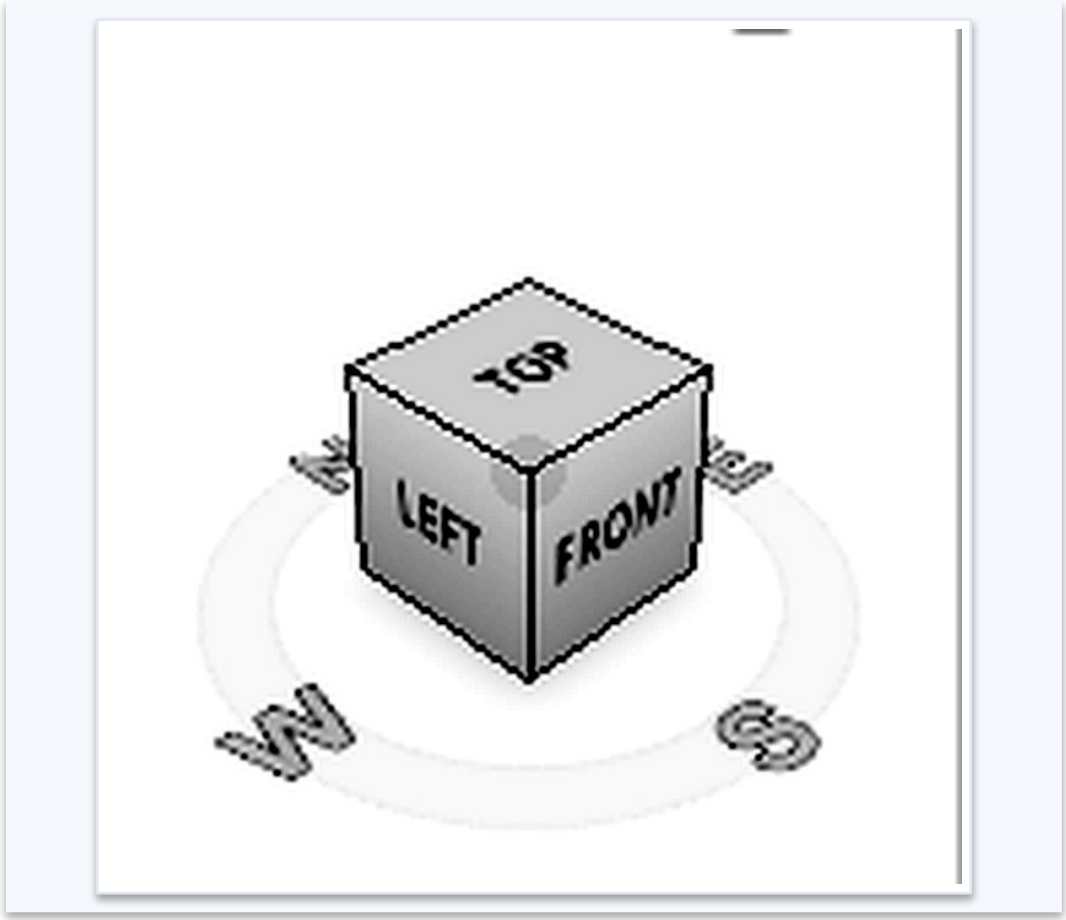


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NAVIGATION

ViewCube and 2D/3D Views

- The ViewCube switches between standard and isometric views of the model.
- Click a face, edge or corner to jump to that view; drag to orbit freely.
- Set Current View as Home to return to a saved orientation.
- In 2D drafting you will mostly stay on the Top view.

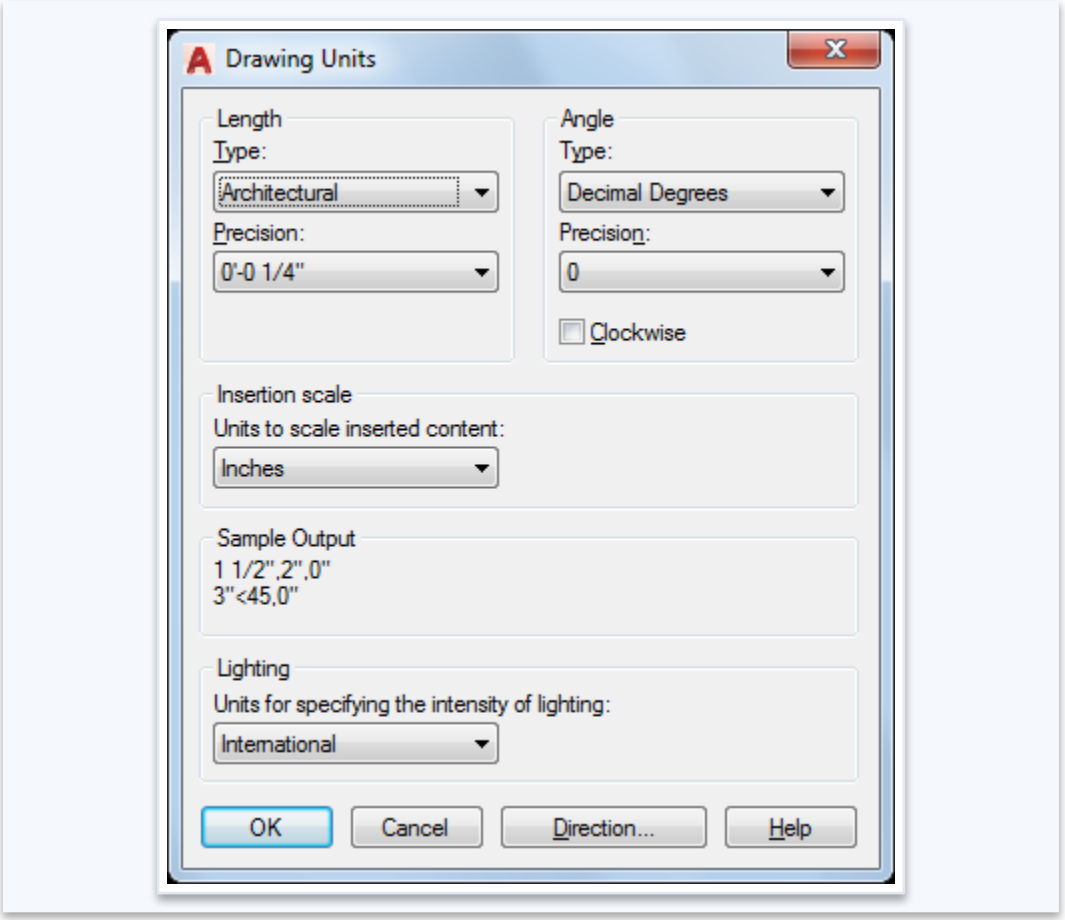


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SETUP

Drawing Units and Converting Units

- UNITS sets the length type (Decimal, Architectural, Engineering…) and precision.
- Angles default to decimal degrees measured anticlockwise from east.
- This course draws in millimetres with decimal units.
- To convert a drawing's scale: SCALE → All → base 0,0 → factor (e.g. 25.4).



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TOPIC 01 · HANDS-ON

Setting Up a New Drawing

LAB 1

Create and verify a metric drawing environment, compare saved workspace and drafting-setting examples, then open a finished drawing and practise reading it with zoom, pan and ViewCube.

You'll produce

A correctly configured metric drawing saved as DWG, a settings audit, and confident navigation of an existing technical drawing.

Commands: ACADISO template, UNITS, LIMITS, DSETTINGS, WSCURRENT, ZOOM, PAN, ViewCube
Files: lab01-table.dwg

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LAB 1 · VERIFY

Setting Up a New Drawing

✅ Check your work

INSUNITS reports millimetres, limits are 420 × 297, Drafting & Annotation is current, the precision-aid audit is complete, and lab01-setup.dwg reopens with the settings intact.

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TOPIC RECAP

Recap — Getting Started with AutoCAD

1

Lab 1 — Setting Up a New Drawing

A correctly configured metric drawing saved as DWG, a settings audit, and confident navigation of an existing technical drawing.

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02

TOPIC 02

Technical Drawing with AutoCAD

Basic AutoCAD Operations · Drawing Objects · Isometric Drawing

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SKILLS FRAMEWORK A2 · K2

Key Concepts — Technical Drawing with AutoCAD

1

Lines & polylines

LINE draws separate segments; PLINE creates one connected object of line and arc segments.

2

Circles & arcs

Circles by centre/radius or diameter; arcs by combinations of start, centre, end, radius and angle.

3

Rectangles & polygons

RECTANG builds closed rectangular polylines; POLYGON draws 3–1024-sided regular polygons.

4

Ellipses, splines & points

Ellipses from axis endpoints; splines as smooth curves through fit points; points as snap nodes.

5

Construction geometry

Construction lines (XLINE) and rays give infinite reference lines for projecting views.

6

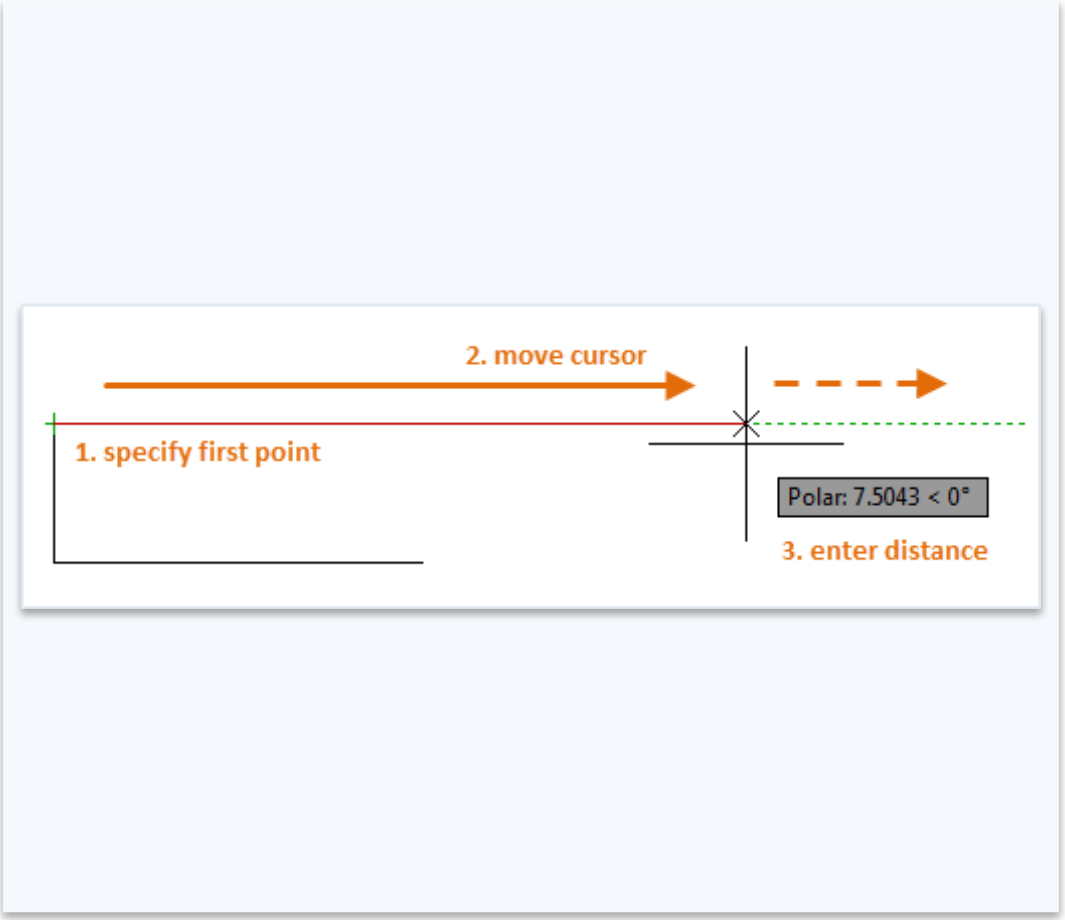
2D isometric drawing

ISODRAFT turns on isometric planes so a flat drawing represents a 3D projection.

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Lines

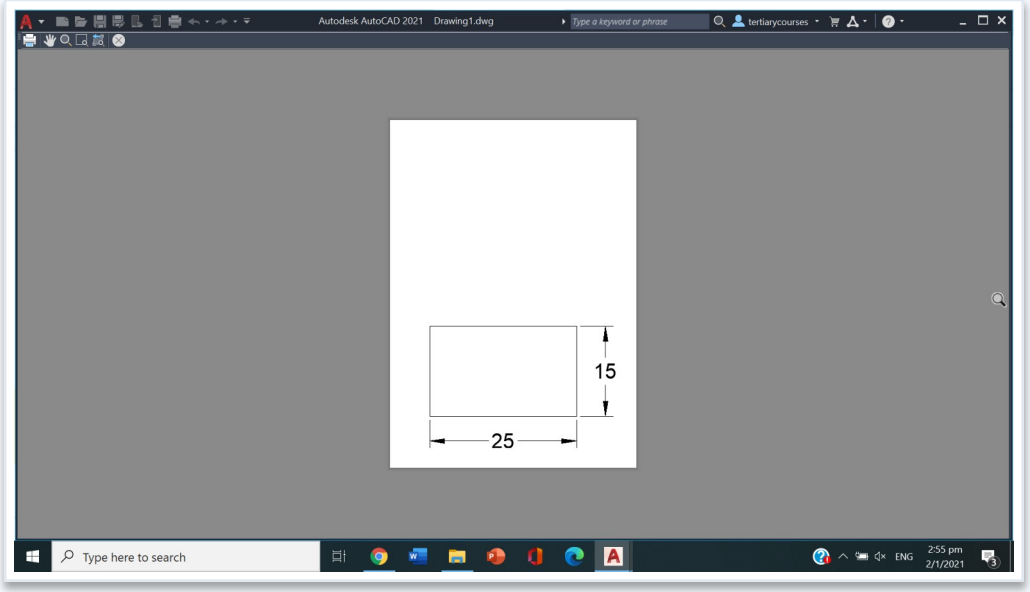
- LINE draws straight segments between picked or typed points.
- Specify the first point, then each next point; every segment is a separate object.
- Combine with object snaps and grid snap for exact geometry.
- Coordinate entry: absolute (100,50) · relative (@50,0) · polar (@50<30).



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Circles and Rectangles

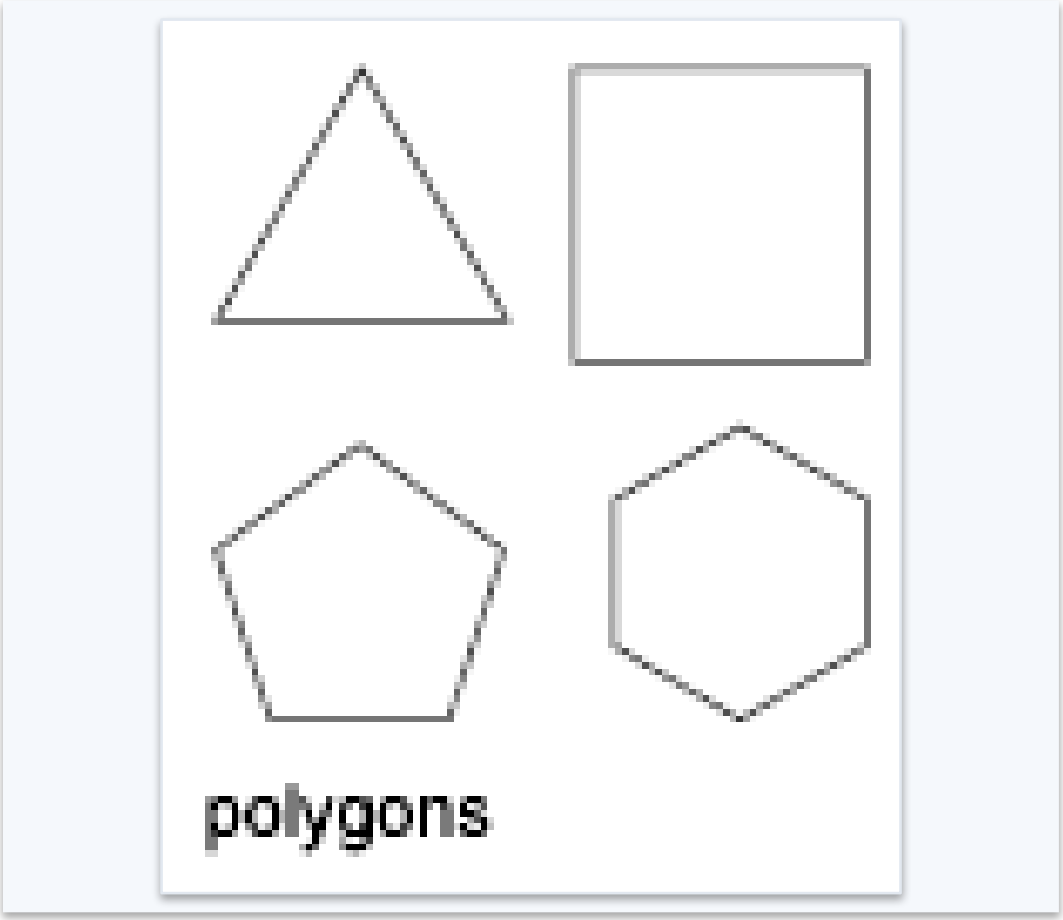
- CIRCLE: centre + radius or diameter; 2P/3P through points; TTR tangent-tangent-radius.
- RECTANG creates a closed rectangular polyline from two corners.
- Rectangle options add chamfered or filleted corners and a set width.
- Use dimensions option (@length,width) for exact rectangle sizes.



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Polygons and Ellipses

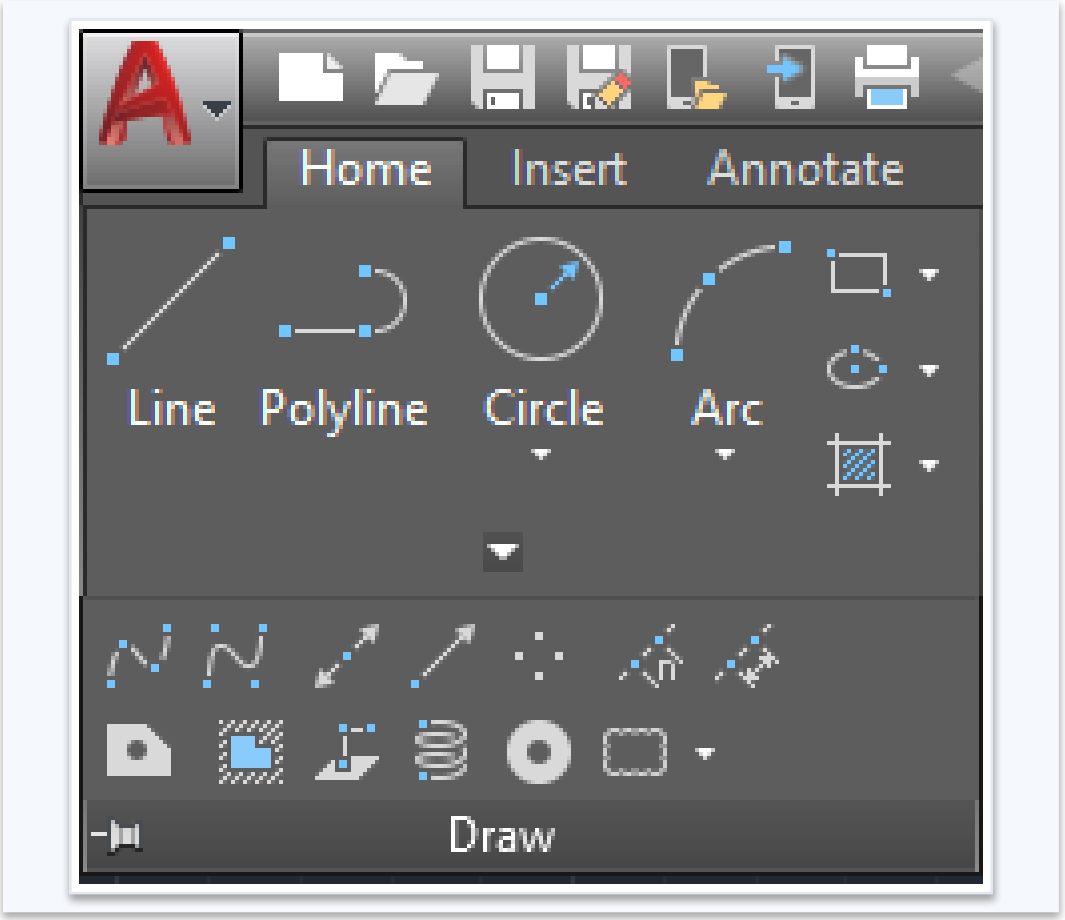
- POLYGON draws regular polygons of 3–1024 sides.
- Inscribed polygons fit inside the reference circle; circumscribed fit outside.
- ELLIPSE uses two axis endpoints then the half-width of the other axis.
- An elliptical arc is an ellipse with start and end angles.



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Arcs, Polylines and Splines

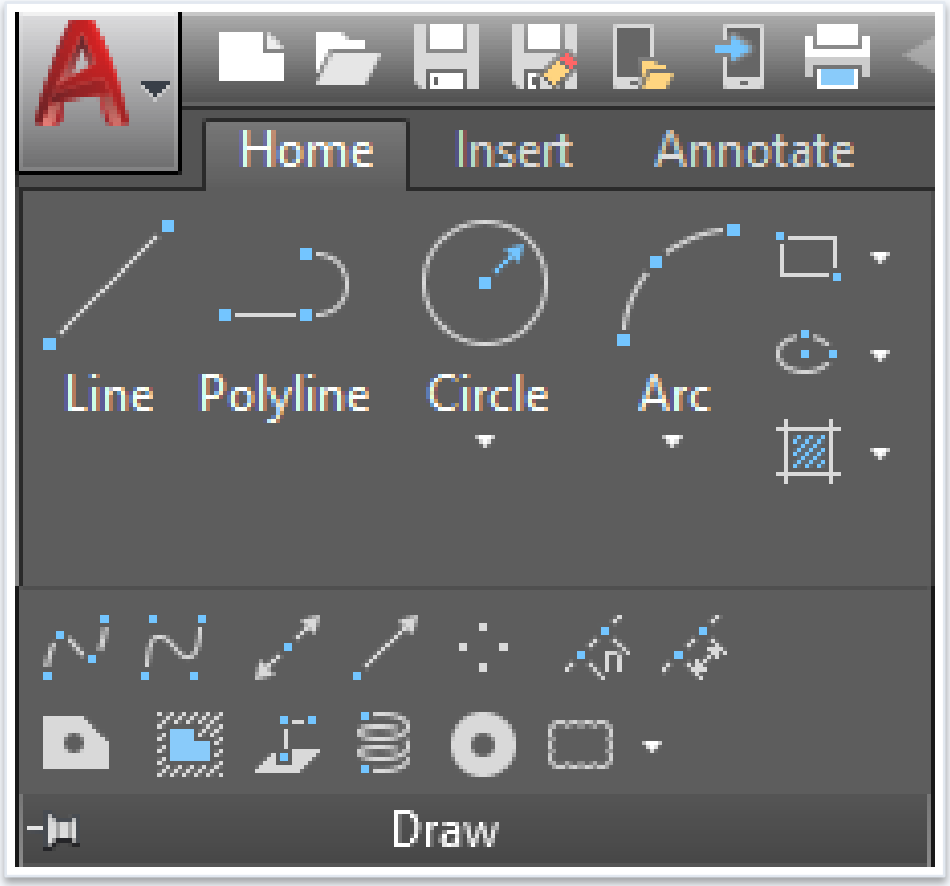
- ARC draws by combinations of start, centre, end, radius, angle and direction.
- PLINE creates one object of connected line and arc segments — switch modes as you draw.
- SPLINE fits a smooth curve through or near a set of fit points.
- Closed polylines are the basis for hatching, areas and 3D operations.



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Points, Construction Lines and Rays

- POINT places node objects; PDMODE and PDSIZE control how they display.
- Points act as snap nodes — DIVIDE and MEASURE place them along objects.
- XLINE creates infinite construction lines for projecting between views.
- RAY starts at a point and extends infinitely in one direction.

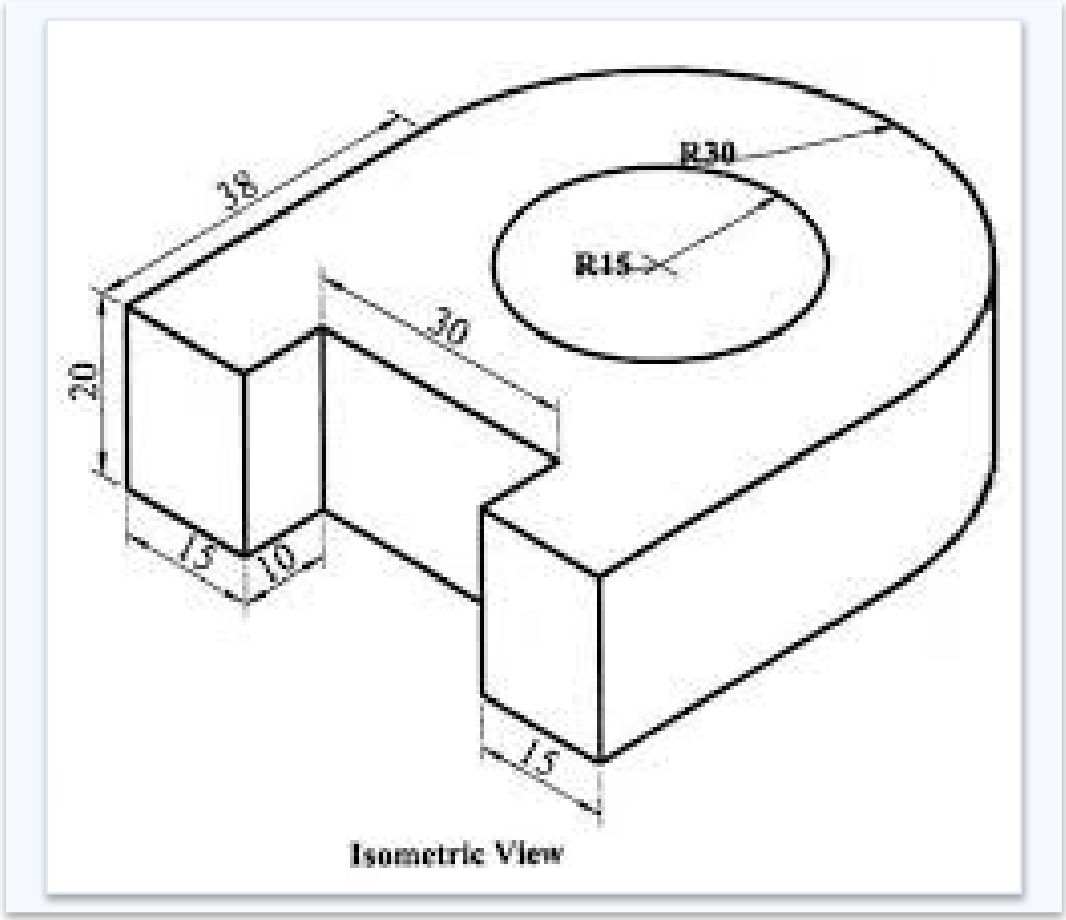


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ISOMETRIC

2D Isometric Drawing with ISODRAFT

- A 2D isometric drawing is a flat representation of a 3D isometric projection.
- ISODRAFT switches the crosshairs to the isometric axes (30°/90°/150°).
- F5 cycles the three isoplanes: Top, Right and Left.
- Isocircles (ELLIPSE → Isocircle) draw round features on the active isoplane.



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WHAT YOU'LL DO

Hands-On Labs — Technical Drawing with AutoCAD

Labs 2–3

- Creating Lines and Coordinates
- Circles, Rectangles and Polygons

Lab 4

- Polylines, Arcs, Ellipses and Splines

Lab 5

- 2D Isometric Drawing

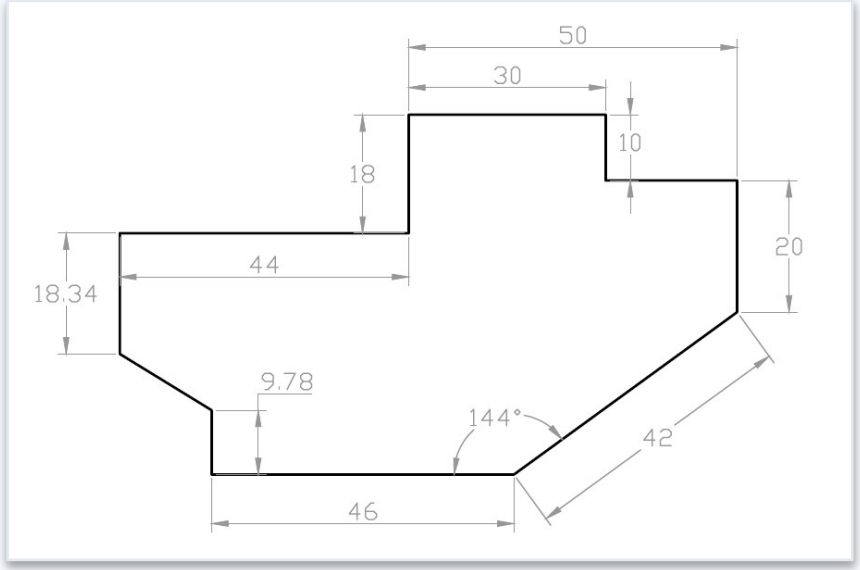
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TOPIC 02 · HANDS-ON

Creating Lines and Coordinates

LAB 2

Draw the outline of practice drawing EX-1 using the LINE and PLINE commands with absolute, relative and polar coordinate entry, working from its dimensioned drawing sheet.



You'll produce

The completed EX-1 profile drawn to size, matching the supplied solution DWG.

Commands: LINE, PLINE, coordinate entry (@dx,dy · @dist<angle), ORTHO, dynamic input

Files: ex1.dwg

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LAB 2 · VERIFY

Creating Lines and Coordinates

✅ Check your work

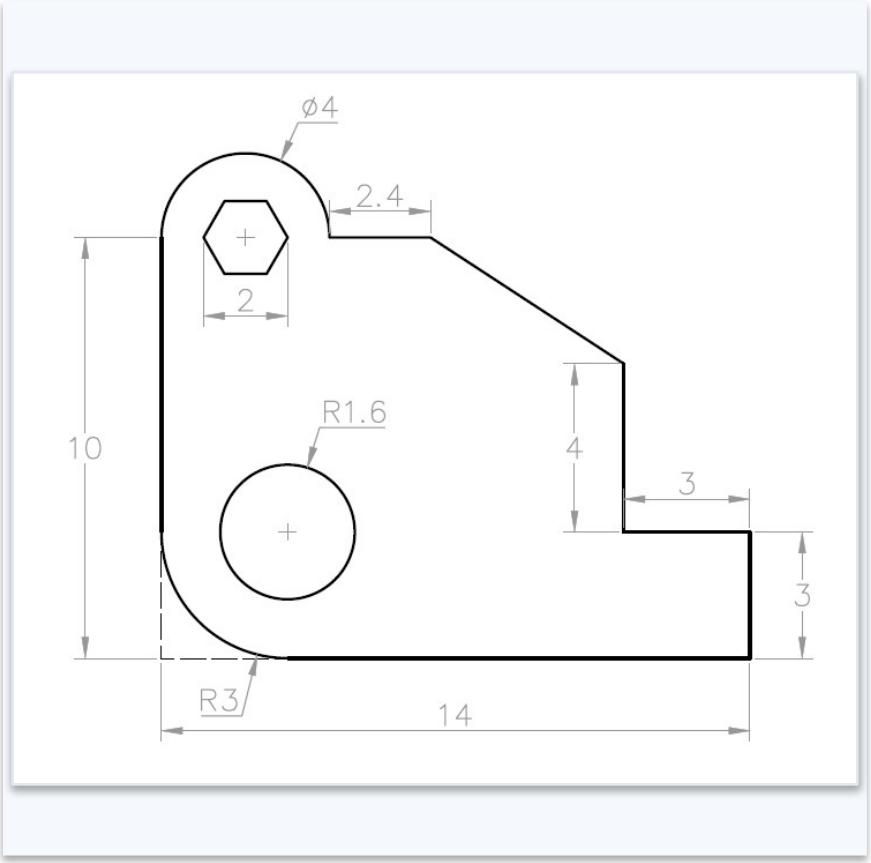
Your outline closes exactly (LIST shows the final vertex at 0,0) and overlays the solution ex1.dwg with no visible deviation.

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Circles, Rectangles and Polygons

LAB 3

Complete practice drawing EX-2 by adding circles (centre-radius, diameter, 2P/3P and Tan-Tan-Radius), a rectangle with chamfered/filleted corners and inscribed/circumscribed polygons.



You'll produce

The completed EX-2 drawing with all holes, rectangles and polygons placed to the sheet dimensions.

Commands: CIRCLE (Cen-Rad, 2P, 3P, TTR), RECTANG, POLYGON, object snaps
Files: ex2.dwg

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LAB 3 · VERIFY

Circles, Rectangles and Polygons

✅ Check your work

Every circle's centre and radius match the sheet (use LIST), the hexagon has 6 equal sides, and your drawing overlays ex2.dwg cleanly.

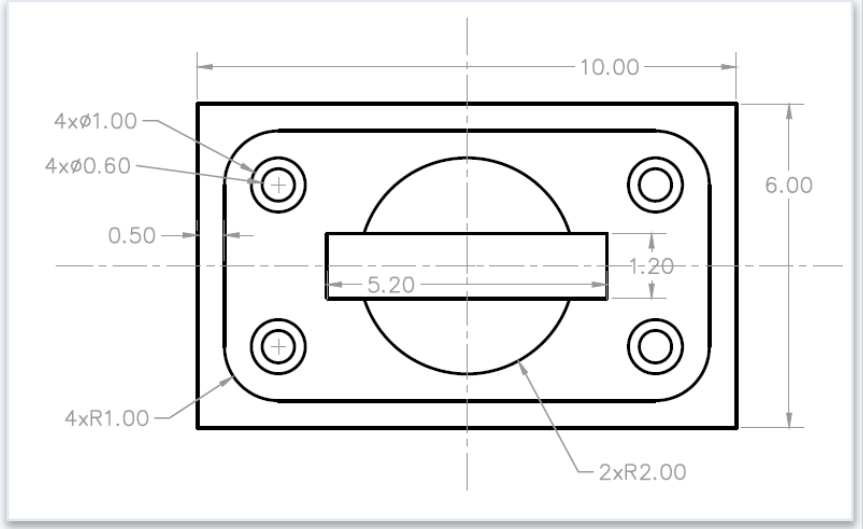
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TOPIC 02 · HANDS-ON

Polylines, Arcs, Ellipses and Splines

LAB 4

Draw practice drawing EX-3 combining polylines with arc segments, standalone arcs, an ellipse and a spline, plus construction lines to project feature positions between views.



You'll produce

The completed EX-3 drawing with mixed line/arc geometry drawn as clean, connected objects.

Commands: PLINE (Arc mode), ARC, ELLIPSE, SPLINE, XLINE, POINT/DIVIDE
Files: ex3.dwg

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LAB 4 · VERIFY

Polylines, Arcs, Ellipses and Splines

✅ Check your work

The outline is a single polyline (one object when selected), arcs and the ellipse match the sheet dimensions, and no construction lines remain in the final drawing.

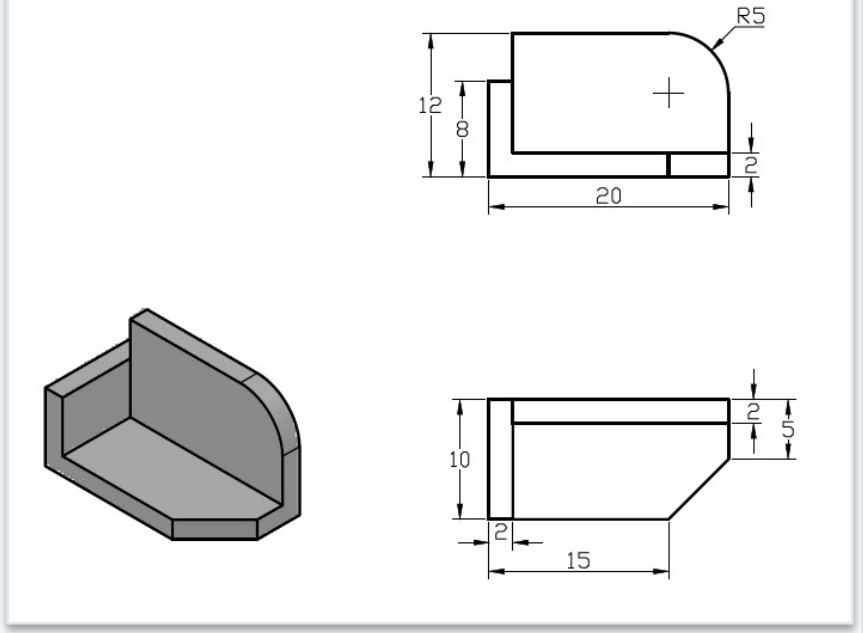
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TOPIC 02 · HANDS-ON

2D Isometric Drawing

LAB 5

Use ISODRAFT to create a 2D isometric representation of the 3D object supplied in iso-object.dwg, switching isoplanes and drawing isocircles for the round features.



You'll produce

A flat isometric drawing of the 3D object — edges on the three isoplanes plus isocircle holes.

Commands: ISODRAFT, isoplanes (F5), ELLIPSE Isocircle, polar tracking
Files: iso-object.dwg

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LAB 5 · VERIFY

2D Isometric Drawing

✅ Check your work

Each face lies on the correct isoplane (edges at 30°/90°/150°), the isocircles read as ellipses aligned to their faces, and the drawing matches the supplied object's proportions.

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TOPIC RECAP

Recap — Technical Drawing with AutoCAD

1

Lab 2 — Creating Lines and Coordinates

The completed EX-1 profile drawn to size, matching the supplied solution DWG.

2

Lab 3 — Circles, Rectangles and Polygons

The completed EX-2 drawing with all holes, rectangles and polygons placed to the sheet dimensions.

3

Lab 4 — Polylines, Arcs, Ellipses and Splines

The completed EX-3 drawing with mixed line/arc geometry drawn as clean, connected objects.

4

Lab 5 — 2D Isometric Drawing

A flat isometric drawing of the 3D object — edges on the three isoplanes plus isocircle holes.

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03

TOPIC 03

Modifying Objects

Move · Copy · Rotate · Offset · Mirror · Trim · Fillet · Arrays

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SKILLS FRAMEWORK A3 · K3

Key Concepts — Modifying Objects

1

Relocate & duplicate

MOVE, COPY and ROTATE reposition objects from a base point to a second point or angle.

2

Offset & mirror

OFFSET creates parallel copies at a set distance; MIRROR reflects half a symmetric design.

3

Trim & extend

Quick and Standard modes cut objects back to, or extend them to meet, boundary edges.

4

Fillet & chamfer

FILLET rounds a corner with a radius; CHAMFER bevels it with distances or an angle.

5

Arrays

Rectangular, polar and path arrays distribute copies in rows/columns, around a centre, or along a curve.

6

Stretch, scale & explode

STRETCH lengthens with a crossing window; SCALE resizes about a base point; EXPLODE breaks compounds.

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MODIFY

Relocating and Duplicating Objects

1

MOVE (M)

Select objects, pick a base point, then the second point — the objects relocate exactly.

2

COPY (CO)

Same flow as MOVE but leaves the original; Multiple mode places copy after copy.

3

ROTATE (RO)

Rotate about a base point by an absolute angle, or use Reference to match geometry.

4

OFFSET (O)

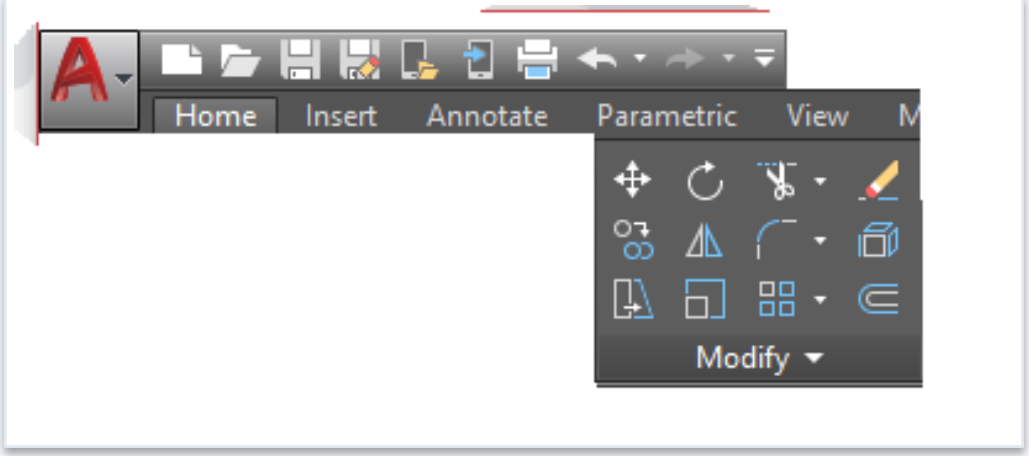
Creates concentric circles, parallel lines and parallel curves at a set distance.

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MODIFY

Offset in Practice

- Specify the offset distance, select the object, then click the side to offset to.
- Through mode offsets through a picked point instead of a distance.
- Offset polylines keep their corners; circles offset concentrically.
- Wall outlines, slots and rims are all one offset away from their centreline.



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MODIFY

Mirror, Stretch and Scale

1

MIRROR (MI)

Draw half of a symmetric design, then reflect it across a two-point mirror line.

2

STRETCH (S)

Select with a crossing window; endpoints inside move, the rest stay anchored.

3

SCALE (SC)

Resize about a base point by factor — or Reference to hit an exact length.

4

Grips

Select first, then drag grips to stretch/move without any command.

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MODIFY

Trim and Extend — Two Modes

Quick mode (default)

- Select the objects to be trimmed or extended, one by one or by dragging
- All visible geometry acts as the cutting/boundary edge
- Fastest for everyday cleanup

Standard mode

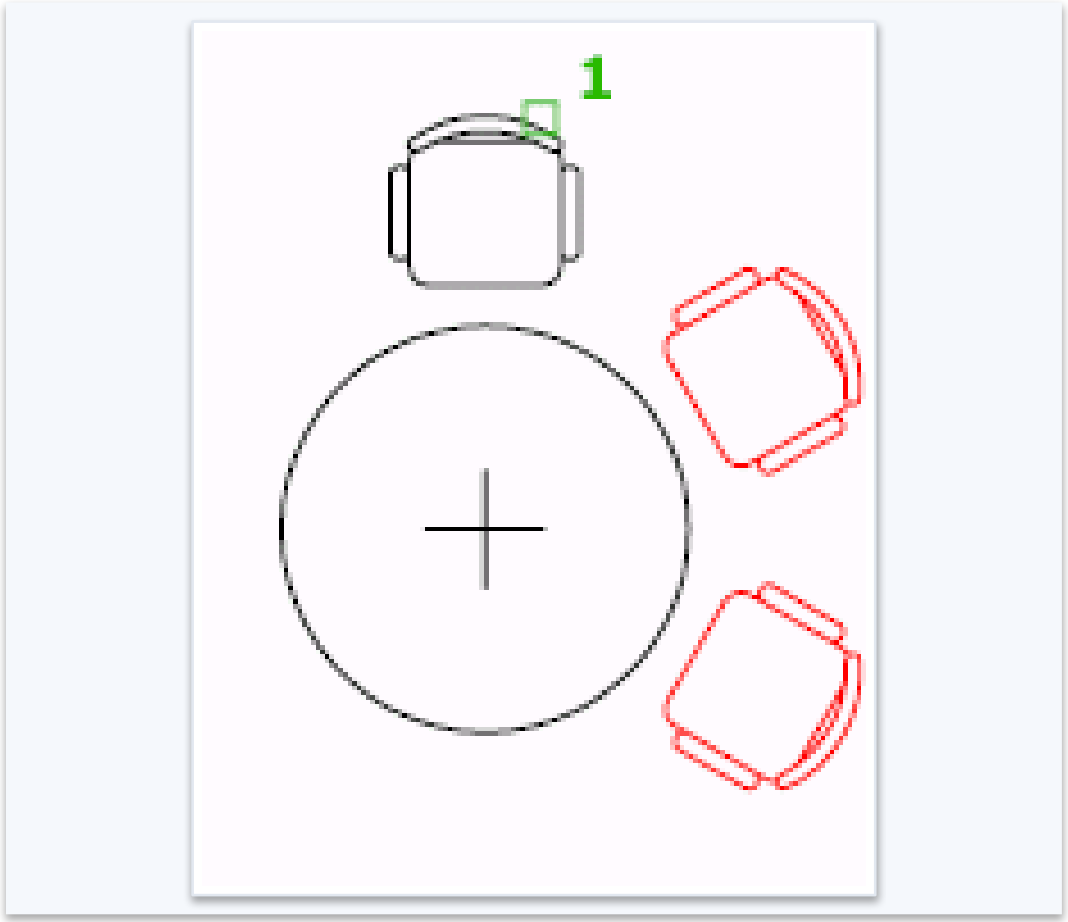
- First select boundary edges, press Enter, then select objects to trim/extend
- Press Enter at the first prompt to make everything a boundary
- Precise control for crowded drawings

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MODIFY

Arrays — Rectangular, Polar and Path

- ARRAY creates copies arranged in a pattern — associative by default.
- Rectangular: rows, columns and levels with set spacing.
- Polar: evenly distributes copies around a centre point or axis.
- Path: distributes copies along a line, polyline or curve.

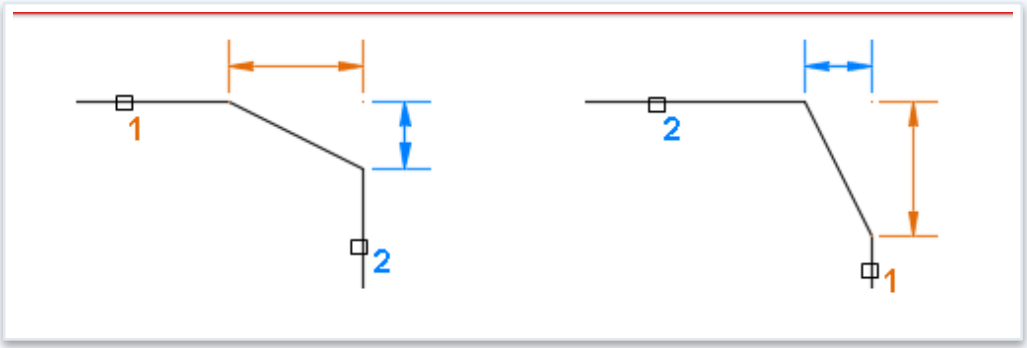


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MODIFY

Fillet and Chamfer

- FILLET rounds the corner between two objects with a set radius.
- Enter r (Radius) first to change the current fillet radius.
- CHAMFER bevels a corner using two distances or a distance and angle.
- Radius 0 / distance 0 are quick ways to square off a corner exactly.



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MODIFY

Erase, Explode, Break, Divide and Measure

1

ERASE (E)

Remove selected objects — or select first and press Delete.

2

EXPLODE (X)

Break a compound object (polyline, block, array) into components.

3

BREAK / Break at Point

Split lines, open polylines and arcs at one or two points.

4

DIVIDE · MEASURE

Place point nodes at equal counts (DIVIDE) or set intervals (MEASURE).

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WHAT YOU'LL DO

Hands-On Labs — Modifying Objects

Lab 6

- Move, Copy, Rotate and Offset

Lab 7

- Trim, Extend, Fillet and Chamfer

Lab 8

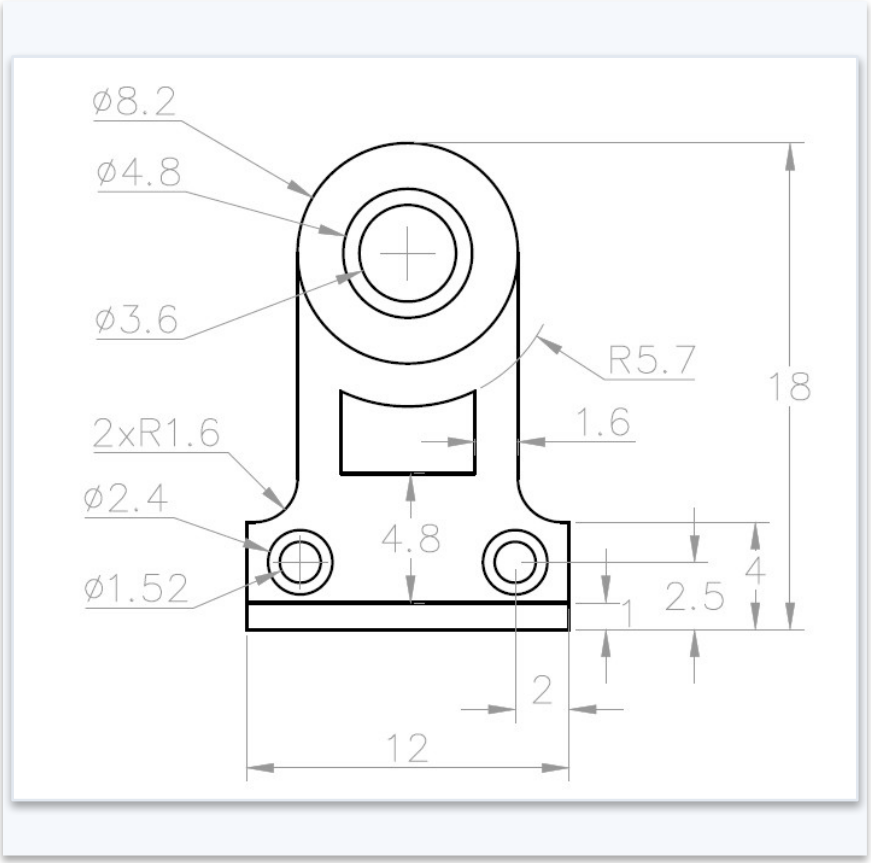
- Mirror, Stretch, Scale and Arrays

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Move, Copy, Rotate and Offset

LAB 6

Build practice drawing EX-4 by drawing one feature and then positioning the rest with MOVE, COPY (multiple), ROTATE with reference angle and OFFSET for the parallel profiles.



You'll produce

The completed EX-4 drawing produced with modify commands rather than re-drawing each feature.

Commands: MOVE, COPY (Multiple/Array), ROTATE (Reference), OFFSET
Files: ex4.dwg

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LAB 6 · VERIFY

Move, Copy, Rotate and Offset

✅ Check your work

Feature spacings measure correct with DIST, offsets are at the specified distances, and the rotated feature sits at exactly the sheet's angle.

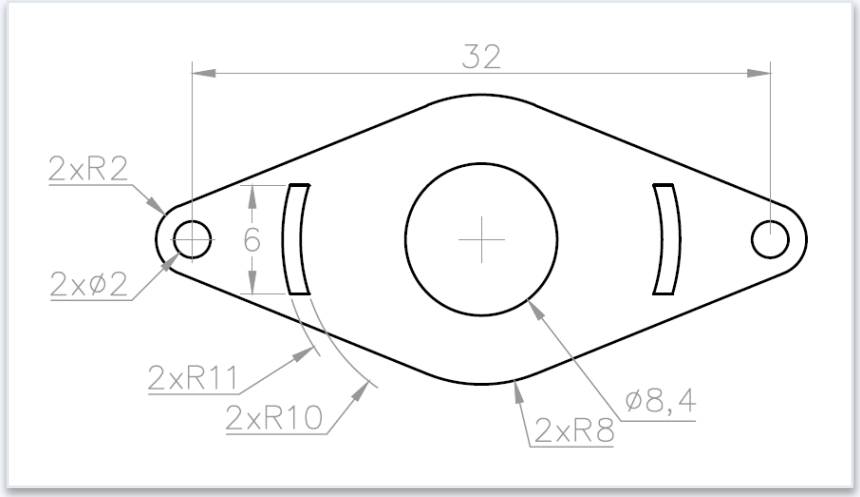
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TOPIC 03 · HANDS-ON

Trim, Extend, Fillet and Chamfer

LAB 7

Complete practice drawing EX-5 by trimming overlapping construction geometry, extending short edges to their boundaries, and finishing corners with fillets and chamfers.



You'll produce

A clean EX-5 profile with every corner finished exactly as dimensioned on the sheet.

Commands: TRIM (Quick/Standard), EXTEND, FILLET, CHAMFER, BREAK
Files: ex5.dwg

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LAB 7 · VERIFY

Trim, Extend, Fillet and Chamfer

✅ Check your work

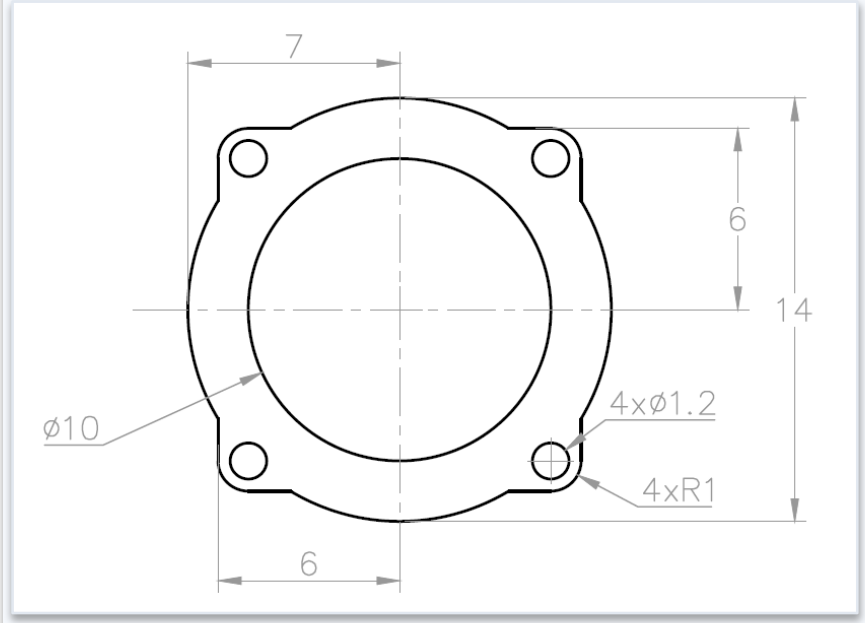
No overshooting or dangling segments remain (zoom around the whole profile), and every fillet radius / chamfer distance matches the drawing sheet.

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Mirror, Stretch, Scale and Arrays

LAB 8

Produce practice drawing EX-6 with mirror, stretch, scale and arrays, then inspect the supplied sawblade drawing to reverse-engineer a production polar-array pattern.



You'll produce

The completed EX-6 drawing plus a verified sawblade polar array with the correct centre, item count, fill angle and associative state.

Commands: MIRROR, STRETCH (crossing window), SCALE (Reference), ARRAYRECT, ARRAYPOLAR

Files: ex6.dwg

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LAB 8 · VERIFY

Mirror, Stretch, Scale and Arrays

✅ Check your work

The two halves are symmetric, array counts and spacings match the sheet, the slot length is correct, and you can explain the four defining properties of the sawblade polar array.

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TOPIC RECAP

Recap — Modifying Objects

1

Lab 6 — Move, Copy, Rotate and Offset

The completed EX-4 drawing produced with modify commands rather than re-drawing each feature.

2

Lab 7 — Trim, Extend, Fillet and Chamfer

A clean EX-5 profile with every corner finished exactly as dimensioned on the sheet.

3

Lab 8 — Mirror, Stretch, Scale and Arrays

The completed EX-6 drawing plus a verified sawblade polar array with the correct centre, item count, fill angle and associative state.

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04

TOPIC 04

Organizing Objects

Object Properties · Layers · Blocks, Attributes & Reusable Content

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SKILLS FRAMEWORK A4 · K4

Key Concepts — Organizing Objects

1

Object properties

Colour, linetype and lineweight — set ByLayer and inspected in the Properties palette (PR).

2

Layers

The primary method of organising a drawing by function; control on/off, freeze/thaw, lock and plot.

3

Grips

Multi-functional grips stretch, move, rotate, scale and copy objects without typing a command.

4

Hide & isolate

Temporarily isolate the objects you are working on and hide the rest of the drawing.

5

Blocks & attributes

Blocks are reusable symbols; attributes attach part numbers and data — the basis of part lists.

6

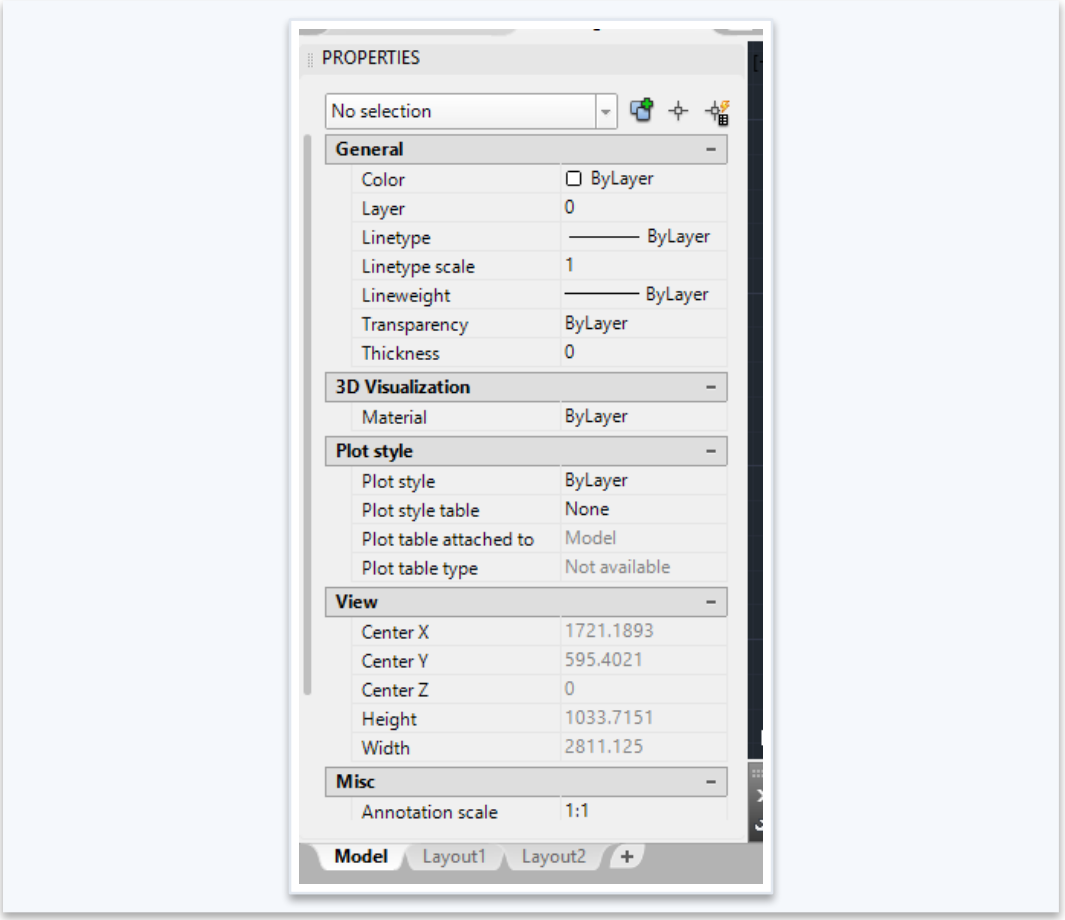
External references

Xrefs attach another drawing (or PDF underlay) so teams share one live reference file.

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Object Properties and the Properties Palette

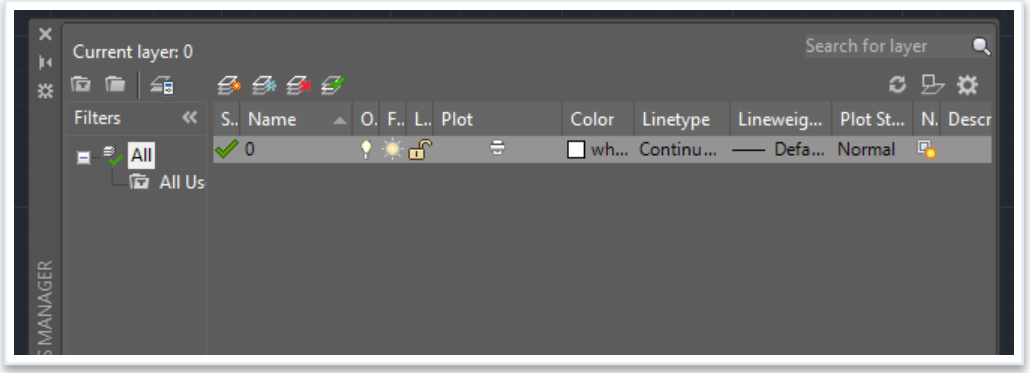
- The Home tab's Layers and Properties panels set the most common properties.
- The Properties palette (PR / Ctrl+1) is the complete list for any selection.
- With nothing selected it shows — and sets — the current defaults.
- Keep colour, linetype and lineweight ByLayer so layers stay in control.



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Layers — Organising Every Drawing

- Layers organise objects by function: outlines, hidden lines, centrelines, text, dimensions.
- The Layer Properties Manager creates layers and sets colour, linetype, lineweight and plot.
- All new objects are drawn on the current layer (green tick).
- Off vs freeze: frozen layers are skipped entirely and speed up big drawings; lock to view but not edit.

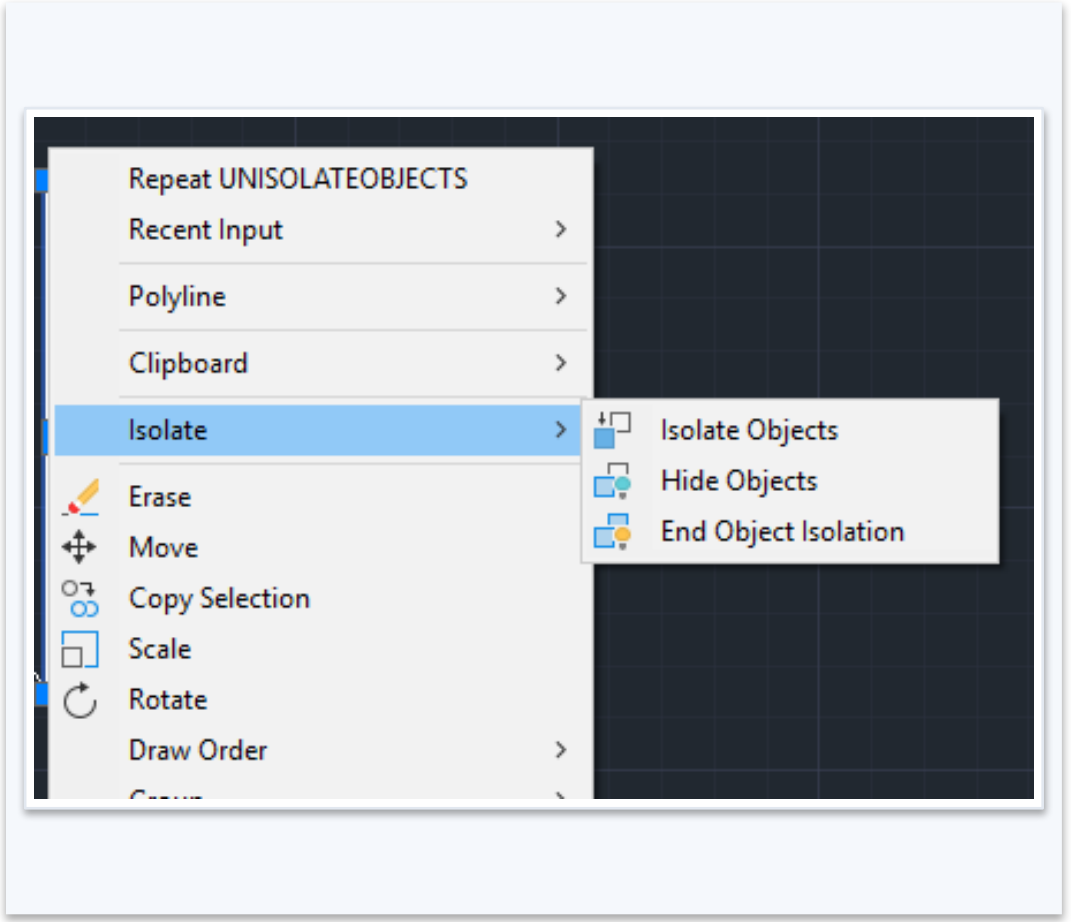


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FOCUS

Hide and Isolate Objects

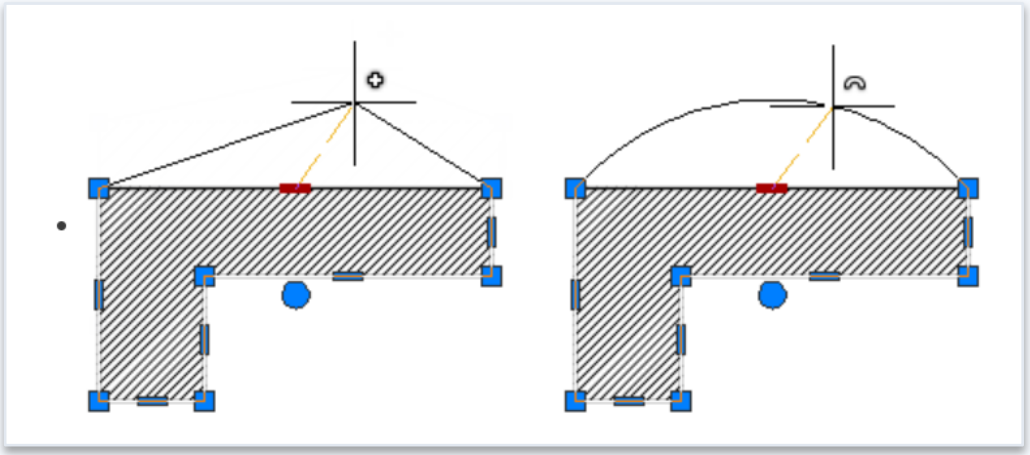
- Isolate Selected Objects hides everything except your selection.
- Hide Selected Objects removes just the selection from view.
- Both are temporary view states — End Object Isolation restores all.
- Right-click menu or the status-bar lightbulb icon controls isolation.



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Multi-Functional Grips

- Select with no command running and grips appear at defining points.
- Drag a grip to stretch; press Enter to cycle Stretch → Move → Rotate → Scale → Mirror.
- Hover a grip for its multi-function menu (e.g. add vertex, convert to arc).
- Lines, polylines, arcs, dimensions and viewports all expose smart grips.

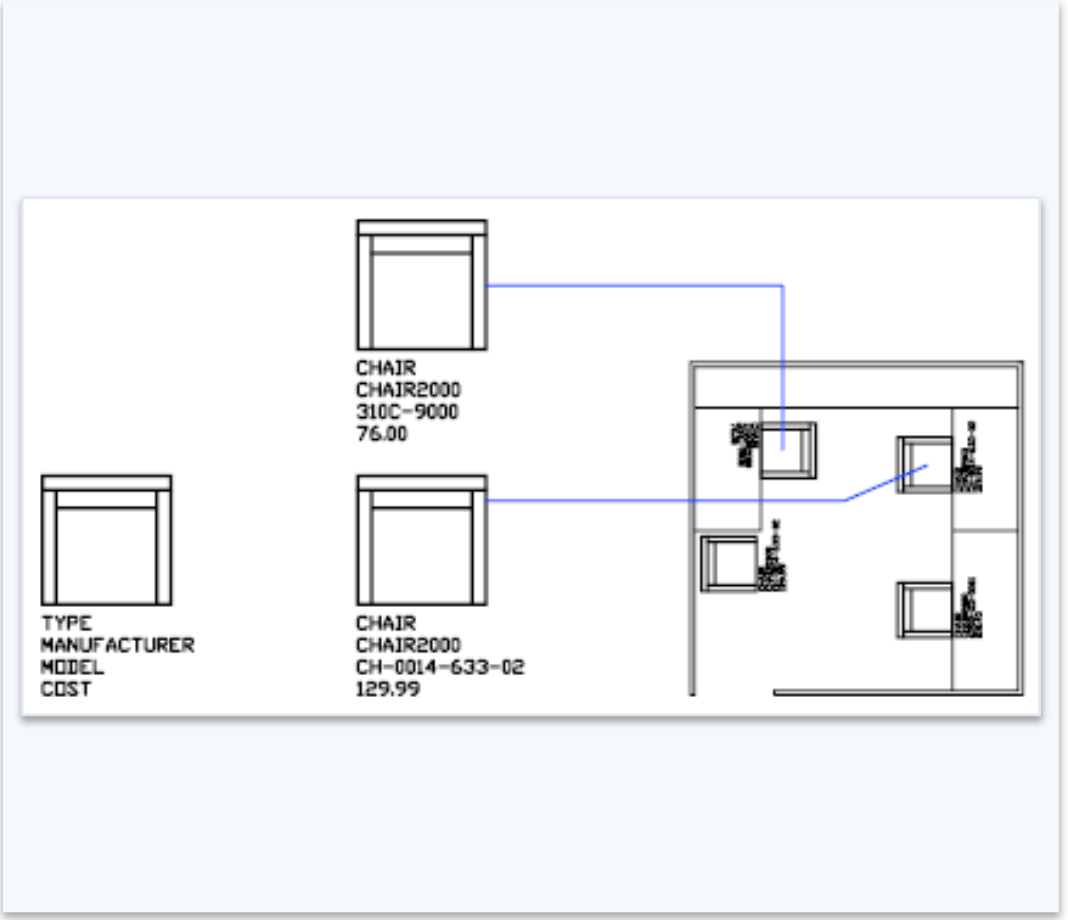


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REUSABLE CONTENT

Blocks, Dynamic Blocks and Attributes

- BLOCK combines objects into a named, reusable symbol inserted with INSERT.
- Dynamic blocks add parameters and actions — one block, many sizes/states.
- Attributes attach data to a block: part numbers, descriptions, quantities.
- ATTDEF defines the tag, prompt and default; DATAEXTRACTION turns attributes into part-list tables.

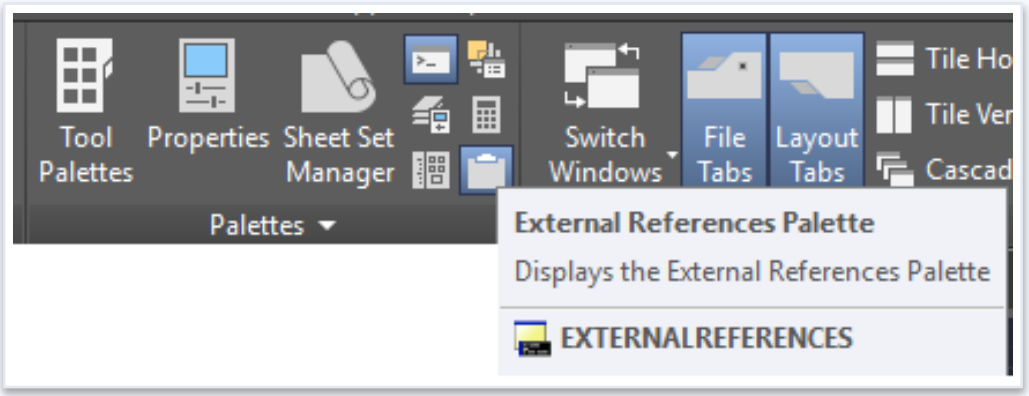


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REUSABLE CONTENT

External References (Xrefs)

- An xref attaches another drawing file into the current drawing — live, not copied.
- The External References palette organises, reloads and repaths every reference.
- Teams share one title block or site plan; edits flow to every host on reload.
- PDFs and images attach the same way as underlays for tracing.



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WHAT YOU'LL DO

Hands-On Labs — Organizing Objects

Labs 9–10

- Object Properties and Layers
- Grips, Hide and Isolate

Lab 11

- Blocks, Attributes and Part Lists

Lab 12

- External References and PDF Import

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TOPIC 04 · HANDS-ON

Object Properties and Layers

LAB 9

Create a standards-based layer scheme in lab-table-lesson2.dwg, then audit the supplied office-plan-ground-floor.dwg to see how a larger architectural drawing separates systems.

You'll produce

A layered table drawing where every object is ByLayer, plus a five-layer audit of the office plan covering naming, colour, linetype, lineweight and plot state.

Commands: LAYER (LA), Properties palette (PR), Layer Walk, MATCHPROP, freeze/thaw, lock
Files: lab-table-lesson2.dwg

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LAB 9 · VERIFY

Object Properties and Layers

✅ Check your work

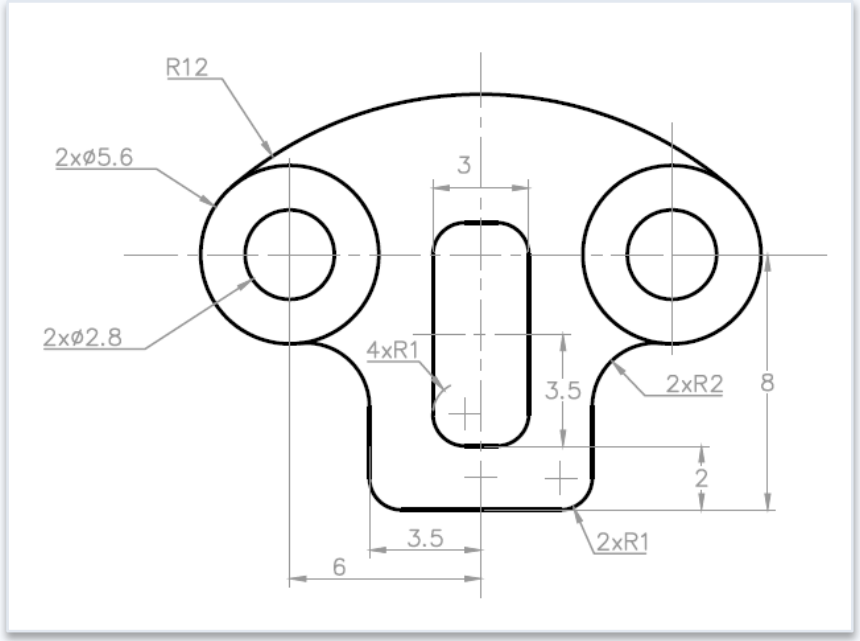
Turning each layer off hides exactly its objects, everything in table-layers.dwg reports ByLayer, and the office-plan audit identifies five layers with their intended functions.

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Grips, Hide and Isolate

LAB 10

Use multi-functional grips on EX-7, then practise isolate, hide, select-similar and end-isolate in the denser new-office-design.dwg without losing or deleting objects.



You'll produce

An edited EX-7 layout plus a saved office-plan view in which the targeted object class was isolated, edited and fully restored.

Commands: Multi-functional grips, grip modes (Enter to cycle), Isolate/Hide Objects, Select Similar

Files: ex7.dwg

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LAB 10 · VERIFY

Grips, Hide and Isolate

✅ Check your work

You can stretch, move, rotate and copy with grips alone; after UNISOLATEOBJECTS, the office drawing's object count and layer visibility match the starting state except for your saved edit.

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TOPIC 04 · HANDS-ON

Blocks, Attributes and Part Lists

LAB 11

Inspect a reusable block library and a drawing containing dynamic blocks, then define a part-tag block with attributes and prepare its data for the part list.

You'll produce

A BALLOON block with PART_NO / DESC / QTY attributes, plus evidence that one library block and one dynamic-block state can be inserted and edited correctly.

Commands: BLOCK, WBLOCK, DesignCenter, ATTDEF, INSERT, ATTEDIT, dynamic-block grips, DATAEXTRACTION
Files: title-block.dwg

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LAB 11 · VERIFY

Blocks, Attributes and Part Lists

✅ Check your work

The library block inserts at the correct scale, the dynamic state changes without exploding, and Data Extraction lists one row per BALLOON with the values you entered.

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TOPIC 04 · HANDS-ON

External References and PDF Import

LAB 12

Repair and verify the relative Landscaping_REV1 Xref in the supplied office-layout host, then attach a title block and a PDF underlay and import vector geometry.

You'll produce

A host drawing whose landscaping Xref resolves from the lab folder, with an additional live title-block Xref and an imported PDF geometry sample.

Commands: XREF/ATTACH, External References palette, PDFATTACH, PDFIMPORT
Files: title-block.dwg

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LAB 12 · VERIFY

External References and PDF Import

✅ Check your work

Both Xrefs show Loaded, Landscaping_REV1 uses a relative path and survives reopening the folder elsewhere, while PDFIMPORT produces editable AutoCAD objects from the underlay.

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TOPIC RECAP

Recap — Organizing Objects

1

Lab 9 — Object Properties and Layers

A layered table drawing where every object is ByLayer, plus a five-layer audit of the office plan covering naming, colour, linetype, lineweight and plot state.

2

Lab 10 — Grips, Hide and Isolate

An edited EX-7 layout plus a saved office-plan view in which the targeted object class was isolated, edited and fully restored.

3

Lab 11 — Blocks, Attributes and Part Lists

A BALLOON block with PART_NO / DESC / QTY attributes, plus evidence that one library block and one dynamic-block state can be inserted and edited correctly.

4

Lab 12 — External References and PDF Import

A host drawing whose landscaping Xref resolves from the lab folder, with an additional live title-block Xref and an imported PDF geometry sample.

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05

TOPIC 05

Design Documentation

Text · Dimensions · Tables & Part Lists · Layouts, Viewports & Plotting

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SKILLS FRAMEWORK A5 · K5

Key Concepts — Design Documentation

1

Text & text styles

Single-line and multiline text; a text style controls font, height and justification centrally.

2

Dimensions

Linear, aligned, angular, radial, diameter, ordinate and arc-length dimensions annotate real sizes.

3

Dimension styles

A named collection of settings — arrowheads, text placement, tolerances — applied consistently.

4

Multileaders & clouds

Multileaders point notes at features; revision clouds flag changed areas for checking.

5

Tables & part lists

TABLE creates row/column data such as bills of materials extracted from block attributes.

6

Layouts & plotting

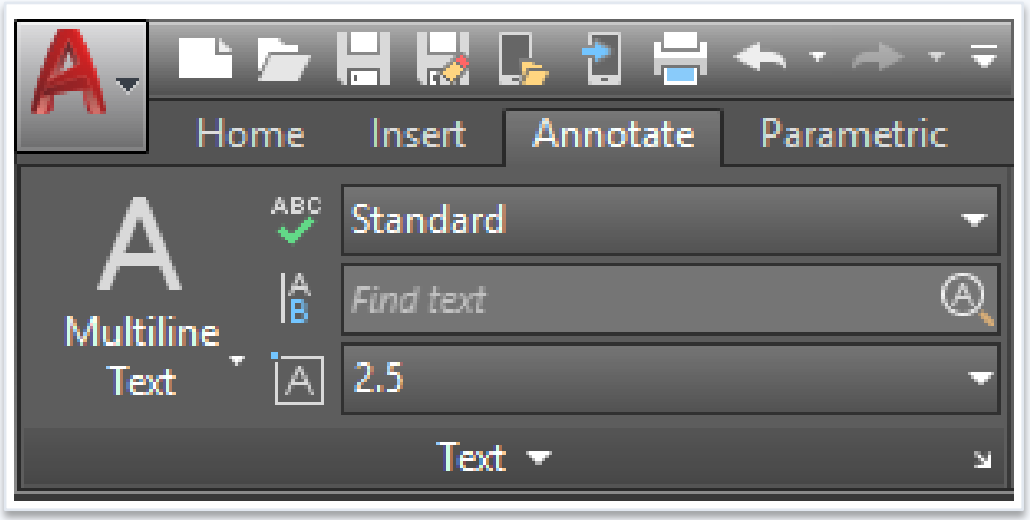
Scaled viewports on a titled layout sheet, printed with PLOT or published to PDF/DWF.

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ANNOTATION

Text and Text Styles

- Multiline text (MTEXT) fills a bounding box; single-line text (TEXT) is quick labels.
- A text style names a collection of settings: font, height, spacing, justification.
- Create styles once (STYLE) and every note updates when the style changes.
- Annotative styles size text automatically to the viewport scale.

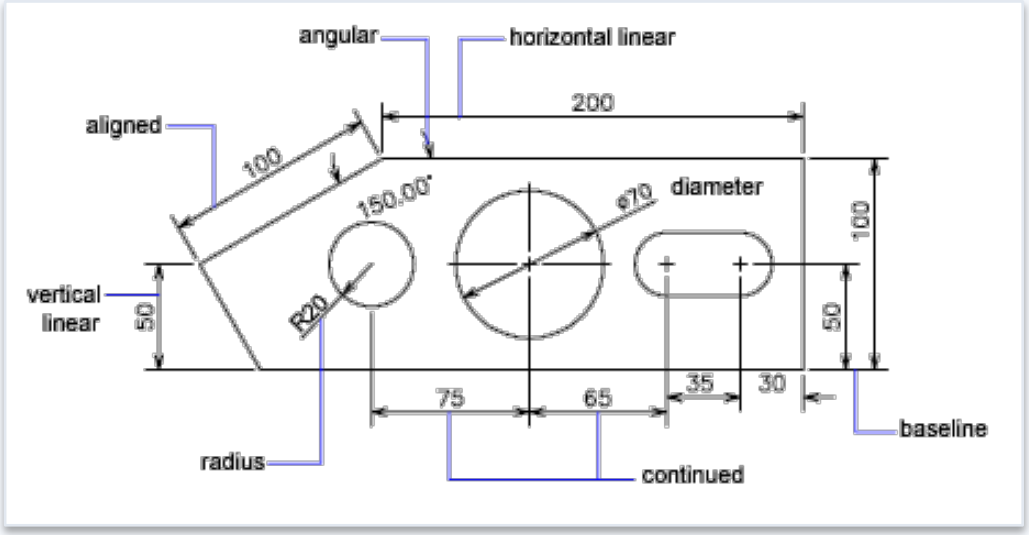


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ANNOTATION

Dimensions — The Basic Types

- Linear and aligned dimensions measure edges; angular measures between lines.
- Radial and diameter dimensions measure arcs and circles, with centre marks.
- Arc-length and ordinate dimensions cover curved lengths and datum offsets.
- Continue chains dimensions end-to-end; Baseline stacks them from one datum.



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ANNOTATION

Dimension Styles

1

What a style controls

Arrowheads, extension lines, text placement, units and tolerances — consistently.

2

Create

DIMSTYLE → New: name it, set fit, text height and precision for your standard.

3

Apply

Set the style current before dimensioning; existing dimensions can be updated.

4

House rule

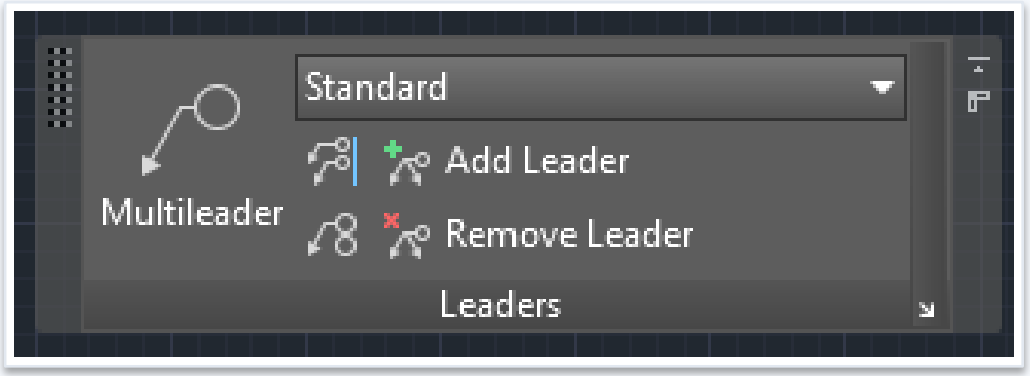
Never override individual dimension text — fix the style instead.

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ANNOTATION

Multileaders and Revision Clouds

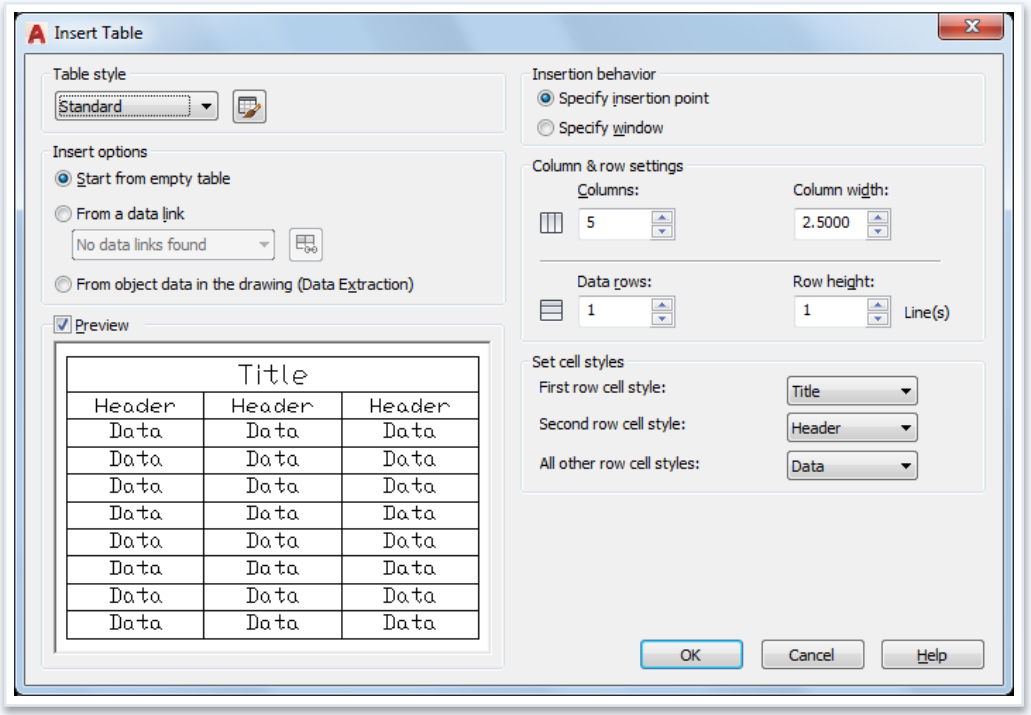
- A multileader is an arrowhead, landing line and note text (or block) in one object.
- MLEADER places it; add/remove leader lines and align multiple leaders neatly.
- Revision clouds flag changed areas for checking and approval.
- Draw a cloud by dragging, corner points — or convert an existing closed object.



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Tables and Part Lists

- TABLE creates a compound object of rows and columns from a table style.
- Type into cells like a spreadsheet; formulas and fields are supported.
- DATAEXTRACTION fills a table from block attributes — the bill of materials.
- Tables can link to external Excel data and update both ways.

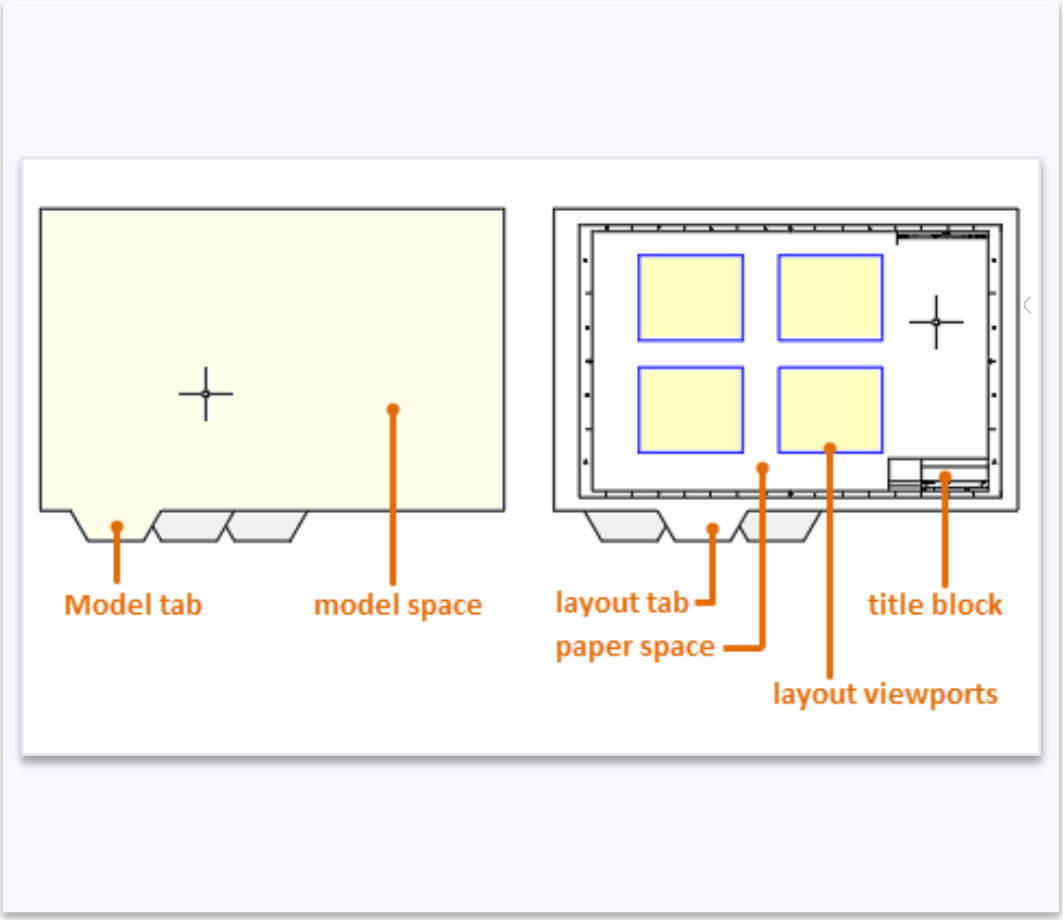


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OUTPUT

Layouts and Viewports

- A layout is the 2D sheet environment: title block, viewports, annotation.
- MVIEW creates layout viewports — windows into model space.
- Set each viewport's scale exactly (e.g. 1:20), then lock it against accidental zooming.
- One model, many sheets: each layout composes its own views and scales.

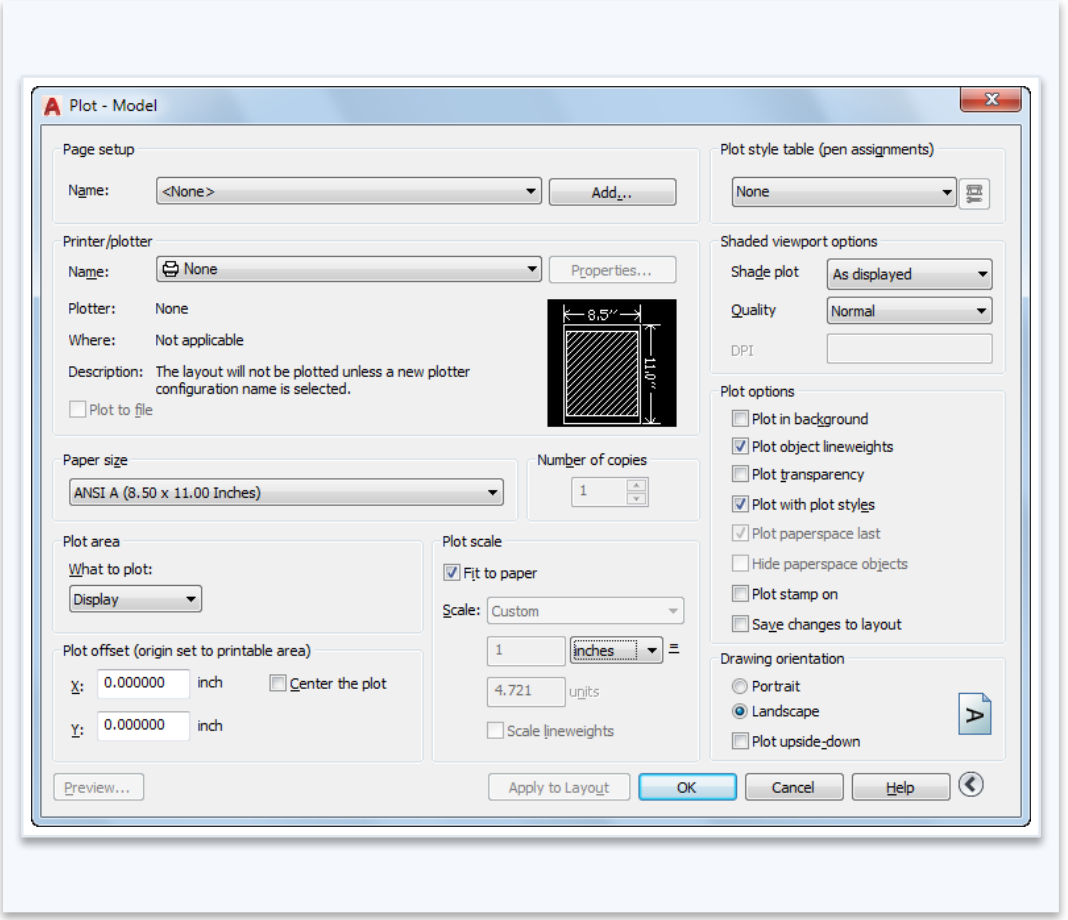


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OUTPUT

Plotting and Printing

- Plot from the layout: choose plotter, paper size, plot area, scale and orientation.
- DWG To PDF.pc3 publishes the standard submission PDF.
- Page setups save the configuration per layout and import between drawings.
- Preview before plotting — check lineweights, scale and the full sheet.

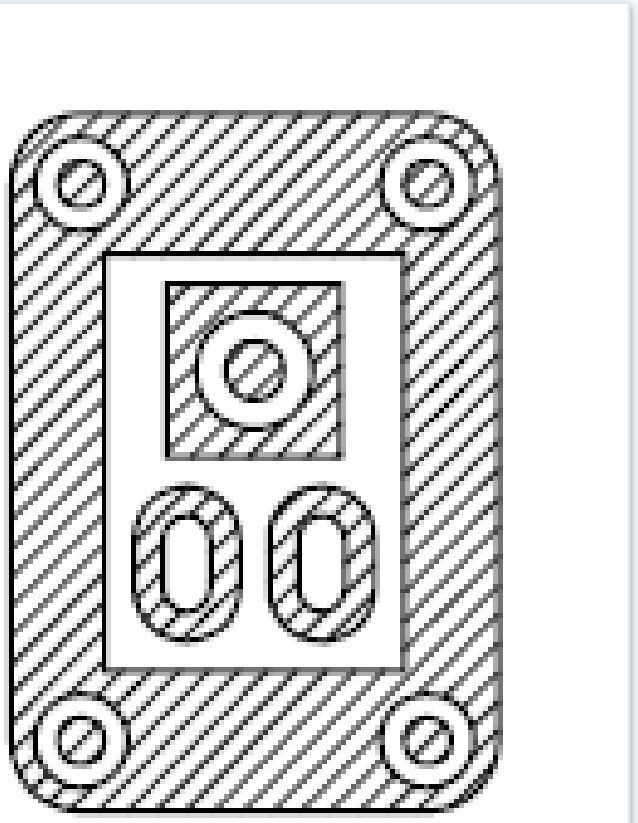


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DETAIL

Hatching and Gradients

- HATCH fills enclosed areas with patterns (ANSI31 for sections), solids or gradients.
- Pick an internal point — island detection keeps text and holes clear.
- Scale and angle make adjacent parts read distinctly in section views.
- HATCHEDIT changes an existing hatch; hatches stay associative to their boundary.



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WHAT YOU'LL DO

Hands-On Labs — Design Documentation

Labs 13–14

- Text and Text Styles
- Dimensions, Multileaders and Part-List Tables

Lab 15

- Hatching and Detail Finishing

Lab 16

- Layouts, Viewports and Plotting

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TOPIC 05 · HANDS-ON

Text and Text Styles

LAB 13

Annotate the layered table drawing with single-line and multiline text, create and apply text styles for titles and notes, and fill in the title-block fields.

You'll produce

An annotated drawing with consistent TITLE and NOTES text styles and a completed title block.

Commands: TEXT, MTEXT, STYLE, annotative text, SPELL
Files: lab-table-lesson2.dwg

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LAB 13 · VERIFY

Text and Text Styles

✅ Check your work

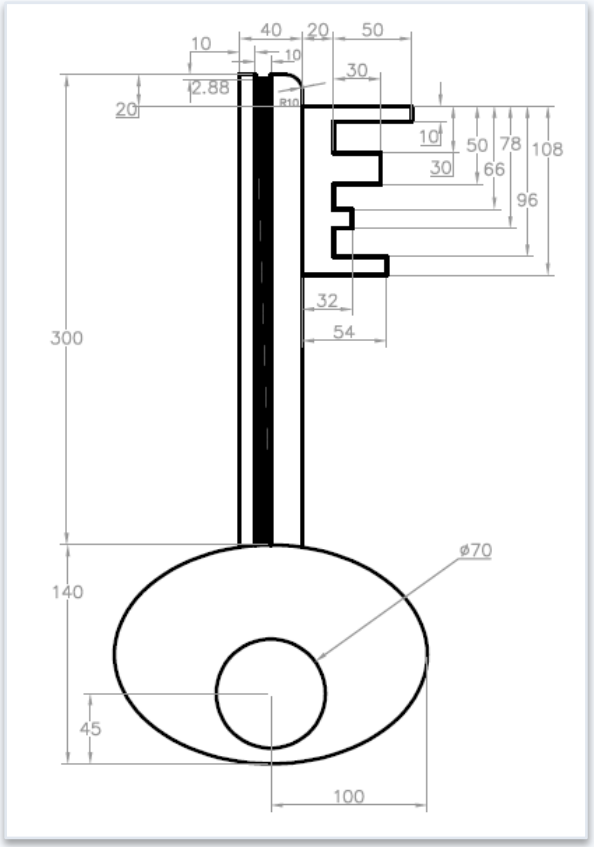
All annotation uses your named styles (none on Standard), the title block is complete, and changing the NOTES style font updates every note at once.

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Dimensions, Multileaders and Part-List Tables

LAB 14

Fully dimension practice drawing EX-8 with a metric dimension style — linear, aligned, radial, diameter, angular, continue and baseline — add multileader callouts, and create the part-list table from the Lab 11 balloon attributes.



You'll produce

A fully dimensioned EX-8 sheet with leader notes and a parts table (part no · description · qty).

Commands: DIMSTYLE, DIM, DLI/DAL/DRA/DDI/DAN, DCO/DBA, MLEADER, TABLE, DATAEXTRACTION

Files: ex8.dwg

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LAB 14 · VERIFY

Dimensions, Multileaders and Part-List Tables

✅ Check your work

Every dimension on the EX-8 sheet is reproduced with the MECH-25 style (no overridden text), leaders land neatly, and the table rows match the balloons one-for-one.

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TOPIC 05 · HANDS-ON

Hatching and Detail Finishing

LAB 15

Apply section hatching with controlled boundaries and islands, then compare two supplied landscaping revisions and mark the changed area with a revision cloud and note.

You'll produce

A correctly hatched section view plus a documented landscaping revision showing the changed geometry, revision cloud and concise change note.

Commands: HATCH, gradient fills, boundary/island detection, HATCHEDIT, DWG Compare, REVCLOUD, MLEADER
Files: ex3.dwg

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LAB 15 · VERIFY

Hatching and Detail Finishing

✅ Check your work

Hatches do not bleed, adjacent parts use distinct angles, islands remain clear, and the landscaping review identifies every revision with one cloud and one readable note.

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TOPIC 05 · HANDS-ON

Layouts, Viewports and Plotting

LAB 16

Audit the supplied new-office-layout.dwg, then compose an A3 layout with a title block, locked scaled viewports and a standards-compliant PDF plot.

You'll produce

A print-ready A3 office layout with two locked viewports, completed title information, verified lineweights and a plotted PDF.

Commands: LAYOUT, PAGESETUP, MVIEW, viewport scale/lock, PLOT, DWG to PDF.pc3
Files: title-block.dwg

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LAB 16 · VERIFY

Layouts, Viewports and Plotting

✅ Check your work

Both viewports report their exact scales in the status bar and cannot be zoomed when locked, and the plotted PDF shows the complete titled sheet at A3.

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TOPIC RECAP

Recap — Design Documentation

1

Lab 13 — Text and Text Styles

An annotated drawing with consistent TITLE and NOTES text styles and a completed title block.

2

Lab 14 — Dimensions, Multileaders and Part-List Tables

A fully dimensioned EX-8 sheet with leader notes and a parts table (part no · description · qty).

3

Lab 15 — Hatching and Detail Finishing

A correctly hatched section view plus a documented landscaping revision showing the changed geometry, revision cloud and concise change note.

4

Lab 16 — Layouts, Viewports and Plotting

A print-ready A3 office layout with two locked viewports, completed title information, verified lineweights and a plotted PDF.

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WRAP-UP

Course Summary & Next Steps

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LEARNING OUTCOMES

What You Achieved

1

Getting started

Navigated the AutoCAD UI, workspaces, units and precision drafting aids.

2

Technical drawing

Produced accurate 2D drawings and isometric views from dimensioned sheets.

3

Modifying objects

Built designs efficiently with move/copy, offset, trim, fillet and arrays.

4

Organizing objects

Structured drawings with layers, blocks, attributes and external references.

5

Design documentation

Dimensioned, annotated, tabulated part lists and plotted submission-ready sheets.

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KEEP LEARNING

Recommended Next Courses

1. **Architecture Drawing with Revit**
   WSQ · Tertiary Infotech Academy

2. **Product Design with Fusion 360**
   WSQ · Tertiary Infotech Academy

3. **AutoCAD Civil 3D for Infrastructure Design**
   WSQ · Tertiary Infotech Academy

4. **Technical Drawing with AutoCAD Mechanical**
   WSQ · Tertiary Infotech Academy

5. **Technical Drawing with AutoCAD Electrical**
   WSQ · Tertiary Infotech Academy

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WE'RE HERE TO HELP

Support

- If you have any enquiries during and after the class, contact us:
- Email: enquiry@tertiaryinfotech.com
- Tel: +65 6100 0613
- Website: www.tertiarycourses.com.sg

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BEFORE YOU LEAVE

Cert & TRAQOM Survey (Mandatory)

- Complete the TRAQOM course-quality survey on the LMS — it is mandatory for WSQ funding.
- Your e-certificate is issued on the LMS after you are assessed Competent and the survey is done.
- Portal: https://lms-tms.tertiaryinfotech.com

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ASSESSMENT

Final Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — produce a dimensioned AutoCAD technical drawing, 1.5 hours, open book.
- Open book: slides, Learner Guide and approved materials only.
- Remember to take the Assessment digital attendance (TRAQOM).
- Submit your completed answers and DWG files on the LMS at https://lms-tms.tertiaryinfotech.com.

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ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit WA (SAQ) then Practical Performance — open book
4. Submit your answers and DWG file on the LMS
5. Sign the Assessment Summary Record

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Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer/administrator displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

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HAPPY DRAFTING

Thank You!

You are now ready to read, produce and document technical drawings with AutoCAD.

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